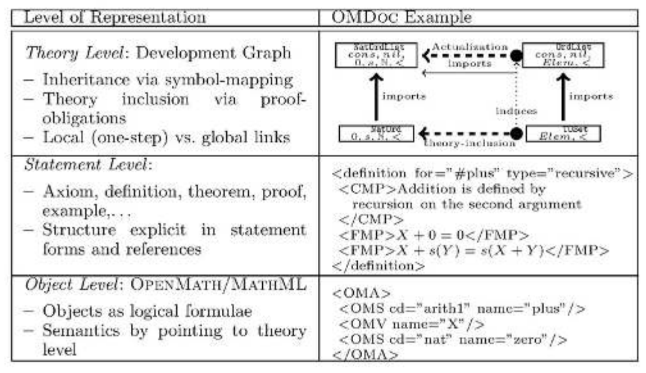
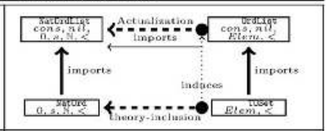
kohlhase-omdoc: formula evidence

The page, the confidence and the picture of an inline formula are its HOST LINE's — a formula has none of its own. A line's confidence is not a formula's.

Inline formulas (first occurrence)

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_F00001	10	1.000	`\Omega`	Ω	tably the ΩMEGA group at Saarland University, the MAYA and ACTIVEMATH
kohlhase-omdoc_F00002	19	0.928	`\mathrm{T}_{\mathrm{E}} \mathrm{X} / \mathrm{LA}_{ \mathrm{E}} \mathrm{X}`	$\mathrm{T}_{\mathrm{E}} \mathrm{X} / \mathrm{LA}_{\mathrm{E}} \mathrm{X}$	the function of identifying the position of the respective section in a sequence
kohlhase-omdoc_F00003	19	1.000	`\mathrm{T}_{\mathrm{E}} \mathrm{X}`	$\mathrm{T}_{\mathrm{E}} \mathrm{X}$	This perceived weakness has lead to markup schemes that concentrate
kohlhase-omdoc_F00004	19	—	`\sqrt\{\sin x\}\`	*(not rendered)*	—
kohlhase-omdoc_F00005	19	0.878	`\sqrt{\sin x}`	$\sqrt{\sin x}$	[Knu84, Lam94] as an example.
kohlhase-omdoc_F00006	21	0.696	`\mathrm{T}_{\mathrm{E}} \mathrm{X} / \mathrm{LaTeX}`	$\mathrm{T}_{\mathrm{E}} \mathrm{X} / \mathrm{LaTeX}$	6 1 Document Markup for the Web
kohlhase-omdoc_F00007	23	0.940	`\langle\langle n s U R I\rangle\rangle`	$\langle\langle n s U R I\rangle\rangle$	
kohlhase-omdoc_F00008	23	0.198	`\langle`	$\langle$	descendants are in the namespace ⟨⟨nsURI⟩⟩, unless they have a namespace
kohlhase-omdoc_F00009	23	0.198	`\rangle=`	$\rangle=$	descendants are in the namespace ⟨⟨nsURI⟩⟩, unless they have a namespace
kohlhase-omdoc_F00010	23	0.198	`\rangle`	$\rangle$	descendants are in the namespace ⟨⟨nsURI⟩⟩, unless they have a namespace
kohlhase-omdoc_F00011	23	0.198	`\langle\langle`	$\langle\langle$	descendants are in the namespace ⟨⟨nsURI⟩⟩, unless they have a namespace
kohlhase-omdoc_F00012	23	0.198	`\rangle\rangle`	$\rangle\rangle$	descendants are in the namespace ⟨⟨nsURI⟩⟩, unless they have a namespace
kohlhase-omdoc_F00013	23	0.190	`n s a)\rangle`	$n s a)\rangle$	an XML simple name, and ⟨⟨nsURI⟩⟩ is the namespace URI. In the scope
kohlhase-omdoc_F00014	23	0.190	`n s a\rangle`	$n s a\rangle$	an XML simple name, and ⟨⟨nsURI⟩⟩ is the namespace URI. In the scope
kohlhase-omdoc_F00015	23	0.917	`\hat{=}`	$\hat{=}$	
kohlhase-omdoc_F00016	25	0.832	`7^{\text {th }}`	$7^{\text {th }}$	⟨⟨uri⟩⟩#⟨⟨id⟩⟩ refers to the unique element in D, that has an attribute of type ID with value ⟨⟨id⟩⟩.
kohlhase-omdoc_F00017	25	0.832	`3^{\text {rd }}`	$3^{\text {rd }}$	⟨⟨uri⟩⟩#⟨⟨id⟩⟩ refers to the unique element in D, that has an attribute of type ID with value ⟨⟨id⟩⟩.
kohlhase-omdoc_F00018	25	0.984	`D`	D	Internet markup languages based on tree structures rather than on sequences
kohlhase-omdoc_F00019	25	0.454	`\langle\langle i d\rangle`	$\langle\langle i d\rangle$	
kohlhase-omdoc_F00020	27	0.990	`\mathrm{T}_{\mathrm{E}} \mathrm{X} / \mathrm{LAT}_{ \mathrm{E}} \mathrm{X}`	$\mathrm{T}_{\mathrm{E}} \mathrm{X} / \mathrm{LAT}_{\mathrm{E}} \mathrm{X}$	Even though mathematic
kohlhase-omdoc_F00021	28	0.348	`{ }^{+}`	${ }^{+}$	ematical objects, such as functions, groups, or differential equations, and also
kohlhase-omdoc_F00022	28	1.000	`{ }^{\text {® }}`	${ }^{\text {®}}$	formulae for the Web are MATHML [ABC+03] and OPENMATH [BCC+04].
kohlhase-omdoc_F00023	29	1.000	`\frac{3}{x+2}`	$\frac{3}{x+2}$	erators (m:mo), bracketed groups (m:mfence), and fractions (m:mfrac); oth-

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_F00024	29	0.991	`\Sigma`	Σ	
kohlhase-omdoc_F00025	29	0.840	`14{ }^{\text {th }}`	$14{ }^{\text {th }}$	
kohlhase-omdoc_F00026	30	0.998	`\mathrm{T}_{\mathrm{E}} \mathrm{X}^{3}`	$\mathrm{T}_{\mathrm{E}} \mathrm{X}^{3}$	
kohlhase-omdoc_F00027	30	1.000	`{ }^{3} \mathrm{~T}_{\mathrm{E}} \mathrm{X}`	${ }^{3} \mathrm{~T}_{\mathrm{E}} \mathrm{X}$	
kohlhase-omdoc_F00028	32	0.992	`\forall a, b . a+b=b+a`	$\forall a, b . a+b=b+a$	<OMS cd="quant1" name="forall"/> <OMBVAR><OMV name="a"/><OMV name="b"/></OMBVAR>
kohlhase-omdoc_F00029	35	1.000	`S=(G, \circ)`	$S=(G, \circ)$	
kohlhase-omdoc_F00030	35	1.000	`e \in G`	$e \in G$	
kohlhase-omdoc_F00031	35	0.988	`e \circ x=x`	$e \circ x=x$	
kohlhase-omdoc_F00032	35	0.988	`x \in G`	$x \in G$	
kohlhase-omdoc_F00033	35	0.988	`e`	e	
kohlhase-omdoc_F00034	35	0.988	`S`	S	
kohlhase-omdoc_F00035	35	0.915	`f`	f	$e \circ x = x$ for all $x \in G$. e is called a left unit of S.
kohlhase-omdoc_F00036	35	—	`\square`	$\square$	—
kohlhase-omdoc_F00037	37	0.491	`\mathrm{BCF}^{+}`	BCF^{+}	
kohlhase-omdoc_F00038	38	0.995	`\left[\mathrm{SBB}^{+} 02\right]`	$\left[\mathrm{SBB}^{+} 02\right]$	planning with the aim of creating a mathematical assistant system.
kohlhase-omdoc_F00039	40	0.772	`1{ }^{\text {st }} 2000`	$1{ }^{\text {st }} 2000$	Version 1.0 of the OMDoc format was released on November 1^{st} 2000 to
kohlhase-omdoc_F00040	40	0.981	`29{ }^{\text {th }} 2001`	$29{ }^{\text {th }} 2001$	into a consistent language format. The changes to version 1.0 were largely
kohlhase-omdoc_F00041 **[conf 0.000]**	41	0.000	`X+0=0</ \mathrm{FMP}>`	$X+0=0</ \mathrm{FMP}>$	Doc uses three levels of modeling corresponding to the concerns raised previously. We have visualized this architecture in Figure 3.1.  Fig. 3.1. OMDoc in a nutshell (the three levels of modeling)

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_F00042 **[conf 0.000]**	41	0.000	`<\mathrm{FMP}>X+s(Y)=s(X+Y)</ \mathrm{FMP}>`	$< \mathrm{FMP} > X+s(Y)=s(X+Y) < / \mathrm{FMP} >$	Doc uses three levels of modeling corresponding to the concerns raised previously. We have visualized this architecture in Figure 3.1. Level of Representation / OMDoc Example *Theory Level*: Development Graph – Inheritance via symbol-mapping – Theory inclusion via proof-obligations – Local (one-step) vs. global links  *Statement Level*: – Axiom, definition, theorem, proof, example,... – Structure explicit in statement forms and references <definition for="#plus" type="recursive"> <CMP>Addition is defined by recursion on the second argument </CMP> <FMP>$X+0=0$</FMP> <FMP>$X+s(Y)=s(X+Y)$</FMP> </definition> *Object Level*: OpenMath/MathML – Objects as logical formulae – Semantics by pointing to theory level <OMA> <OMS cd="arith1" name="plus"/> <OMV name="X"/> <OMS cd="nat" name="zero"/> </OMA> **Fig. 3.1.** OMDoc in a nutshell (the three levels of modeling)
kohlhase-omdoc_F00043	42	0.712	`{ }^{+} 04`	${ }^{+} 04$	
kohlhase-omdoc_F00044	43	0.501	`{ }^{\text {TM }}`	${ }^{\text {TM }}$	
kohlhase-omdoc_F00045	43	0.820	`\left[\mathrm{CGG}^{+} 92\right]`	$\left[\mathrm{CGG}^{+} 92\right]$	
kohlhase-omdoc_F00046	48	0.988	`\mathrm{LAT}_{\mathrm{E}} \mathrm{X}`	$\mathrm{LAT}_{\mathrm{E}} \mathrm{X}$	format in mathematics nowadays.
kohlhase-omdoc_F00047	49	0.851	`E`	E	
kohlhase-omdoc_F00048	49	0.851	`E \times E`	$E \times E$	
kohlhase-omdoc_F00049	49	0.988	`f(x, y)`	$f(x, y)$	36 4 Textbooks and Articles
kohlhase-omdoc_F00050	49	0.988	`(x, y) \in E \times E`	$(x, y) \in E \times E$	36 4 Textbooks and Articles
kohlhase-omdoc_F00051	49	0.965	`x`	x	
kohlhase-omdoc_F00052	49	0.965	`y`	y	
kohlhase-omdoc_F00053	49	0.999	`x+y, x . y`	$x+y, x . y$	order and separating them by a characteristic symbol of the law in question (a
kohlhase-omdoc_F00054	49	0.999	`x y`	$x y$	order and separating them by a characteristic symbol of the law in question (a
kohlhase-omdoc_F00055	49	1.000	`x+y`	$x+y$	symbols the composition of x and y is written respectively as $x+y$, $x.y$ or xy.
kohlhase-omdoc_F00056	49	0.999	`x . y=x y`	$x . y=x y$	$x+y$ being called the *sum* of x and y) and we say that it is *written additively*; a law denoted by the symbol . is usually called *multiplication* (the composition
kohlhase-omdoc_F00057	49	0.964	`\top`	$\top$	use the symbols $\top$ and $\perp$ to denote arbitrary laws of composition.
kohlhase-omdoc_F00058	49	1.000	`(X, Y) \mapsto X \cup Y`	$(X, Y) \mapsto X \cup Y$	laws of composition on the set of subsets of a set E.
kohlhase-omdoc_F00059	49	1.000	`(X, Y) \mapsto X \cap Y`	$(X, Y) \mapsto X \cap Y$	laws of composition on the set of subsets of a set E.
kohlhase-omdoc_F00060	49	1.000	`x \in \mathbf{N}`	$x \in \mathbf{N}$	*Theory*, III, §3, no. 4).
kohlhase-omdoc_F00061	49	1.000	`y \in \mathbf{N}`	$y \in \mathbf{N}$	*Theory*, III, §3, no. 4).
kohlhase-omdoc_F00062	49	0.997	`x+y, x y`	$x+y, x y$	on the set of subsets of $E \times E$ (*Set Theory*, II, §3, no. 3, Definition 6); the
kohlhase-omdoc_F00063	49	0.997	`x . y`	$x . y$	on the set of subsets of $E \times E$ (*Set Theory*, II, §3, no. 3, Definition 6); the

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_F00064	49	0.997	`x^{y}`	x^{y}	on the set of subsets of $E \times E$ (*Set Theory*, II, §3, no. 3, Definition 6); the
kohlhase-omdoc_F00065	49	1.000	`(X, Y) \mapsto X \circ Y`	$(X, Y) \mapsto X \circ Y$	into E (*Set Theory*, II, §5, no. 2).
kohlhase-omdoc_F00066	49	1.000	`(f, g) \mapsto f \circ g`	$(f, g) \mapsto f \circ g$	**Fig. 4.1.** A fragment from Bourbaki's algebra [Bou74]
kohlhase-omdoc_F00067	52	0.221	`\mathrm{APT}_{\mathrm{E}} \mathrm{X}`	$\mathrm{APT}_{\mathrm{E}} \mathrm{X}$	4.2 Structure and Statements 39 24,25 | The `cc:permissions` element gives the world the permission to reproduce and distribute it freely. Furthermore the license grants the public the right to make derivative works under certain conditions. | 12.3 p. 103 26 | The `cc:prohibitions` can be used to prohibit certain uses of the document, but this one is unencumbered. | 12.3 p. 103 27 | The `cc:requirements` states conditions under which the license is granted. In our case the licensee is required to keep the copyright notice and license notices intact during distribution, to give credit to the copyright holder, and that any derivative works derived from this document must be licensed under the same terms as this document (the copyleft clause). | 12.3 p. 103 31-37 | The `omtext` element is used to mark up text fragments. Here, we have simply used a single `omtext` to classify the whole text in the fragment as unspecific "text". | 14.3 p. 124 32-36 | The `CMP` element holds the actual text in a multilingual group. Its `xml:lang` specifies the language. If the document is used with a DTD or an XML schema (as we are) this attribute is redundant, since the default value given by the DTD or schema is `en`. More keywords in other languages can be given by adding more `CMP` elements. | 14.1 p. 122 33-35 | The text of the LaTeX fragment we are migrating. For simplicity we do not change the text, and leave that to later stages of the migration. | 38 | The closing tag of the root `omdoc` element. There may not be text after this in the file. | 11.1 p. 90
kohlhase-omdoc_F00068	59	0.999	`\lambda`	λ	formal definition of the property of being a law of composition in the form of
kohlhase-omdoc_F00069	59	0.999	`\lambda E, F . \operatorname{set}(E) \wedge F: E \times E \rightarrow E^{2}`	$\lambda E, F . \operatorname{set}(E) \wedge F: E \times E \rightarrow E^{2}$	formal definition of the property of being a law of composition in the form of
kohlhase-omdoc_F00070	59	1.000	`(A, B)`	(A, B)	it occurs. So if we have `law_of_composition`(A, B) somewhere this can be
kohlhase-omdoc_F00071	59	0.863	`(\lambda E, F . \operatorname{set}(E) \wedge F: E \times E \rightarrow E)(A, B)`	$(\lambda E, F . \operatorname{set}(E) \wedge F: E \times E \rightarrow E)(A, B)$	reduced to $(\lambda E, F.set(E) \wedge F : E \times E \rightarrow E)(A, B)$ which in turn reduces to $set(A) \wedge B : A \times A \rightarrow A$ or in other words `law_of_composition`(A, B) is
kohlhase-omdoc_F00072	59	0.917	`\operatorname{set}(A) \wedge B: A \times A \rightarrow A`	$\operatorname{set}(A) \wedge B: A \times A \rightarrow A$	true, iff A is a set and B is a function from $A \times A$ to A. This definition is
kohlhase-omdoc_F00073	59	1.000	`A`	A	directly used in the second formal definition, which we depict in Listing 4.8.
kohlhase-omdoc_F00074	59	1.000	`B`	B	directly used in the second formal definition, which we depict in Listing 4.8.
kohlhase-omdoc_F00075	59	1.000	`A \times A`	$A \times A$	directly used in the second formal definition, which we depict in Listing 4.8.
kohlhase-omdoc_F00076	59	0.819	`\lambda X . A`	$\lambda X . A$	
kohlhase-omdoc_F00077	59	1.000	`X`	X	
kohlhase-omdoc_F00078	60	0.967	`M`	M	in other words: M is a magma, iff it is a pair (E, F), where F is a law of
kohlhase-omdoc_F00079	60	0.967	`(E, F)`	(E, F)	in other words: M is a magma, iff it is a pair (E, F), where F is a law of
kohlhase-omdoc_F00080	60	0.967	`F`	F	in other words: M is a magma, iff it is a pair (E, F), where F is a law of
kohlhase-omdoc_F00081	62	0.986	`\left[\mathrm{BCC}^{+} 04\right]`	$\left[\mathrm{BCC}^{+} 04\right]$	Content Dictionaries are structured documents used by the OPENMATH stan- dard [BCC+04] to codify knowledge about mathematical symbols and con-

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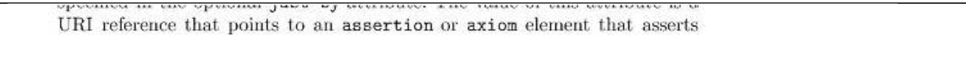
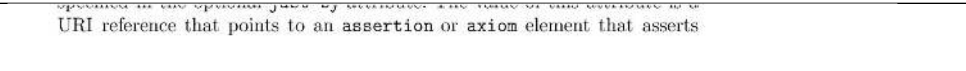
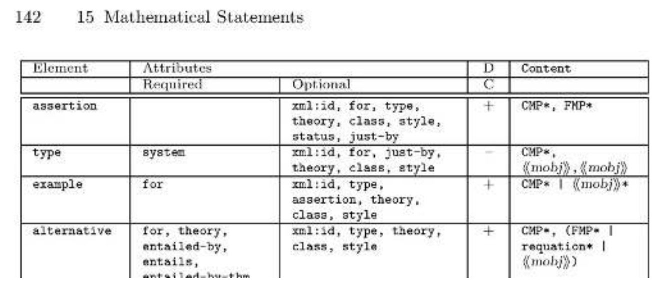
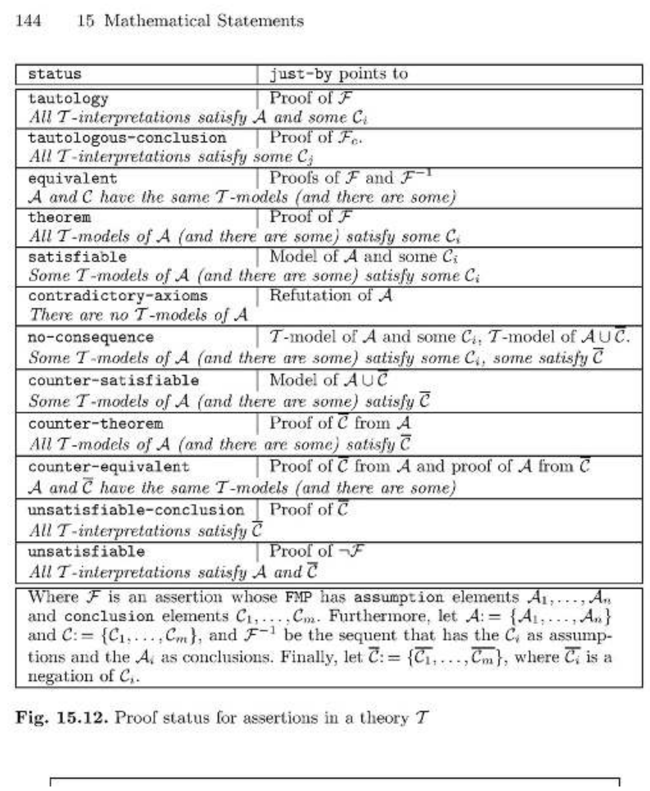
kohlhase-omdoc_F00082	66	1.000	`n`	n	January 17, 2004
kohlhase-omdoc_F00083	66	0.981	`\operatorname{lcm}(a, b)`	$\operatorname{lcm}(a, b)$	The symbol to represent the n-ary tiple of its arguments.
kohlhase-omdoc_F00084	66	0.981	`\frac{a \cdot b}{\operatorname{gcd}(a, b)}`	$\frac{a \cdot b}{\operatorname{gcd}(a, b)}$	The symbol to represent the n-ary tiple of its arguments.
kohlhase-omdoc_F00085	66	0.940	`a`	a	**Definition 2.**(lcm-def)
kohlhase-omdoc_F00086	66	0.940	`b`	b	**Definition 2.**(lcm-def)
kohlhase-omdoc_F00087	66	0.940	`c>0`	$c>0$	**Definition 2.**(lcm-def)
kohlhase-omdoc_F00088	66	0.940	`(a \mid c)`	$(a \mid c)$	**Definition 2.**(lcm-def)
kohlhase-omdoc_F00089	66	0.940	`(b \mid c)`	$(b \mid c)$	**Definition 2.**(lcm-def)
kohlhase-omdoc_F00090	66	1.000	`c<\operatorname{lcm}(a, b)`	$c<\operatorname{lcm}(a, b)$	We define
kohlhase-omdoc_F00091	66	0.998	`k g V(a, b)`	$k g V(a, b)$	Fig. 5.4 A human oriented presentation in
kohlhase-omdoc_F00092	66	0.998	`\frac{a \cdot b}{g g T(a, b)}`	$\frac{a \cdot b}{g g T(a, b)}$	Fig. 5.4 A human oriented presentation in
kohlhase-omdoc_F00093	66	0.354	`b \mid c`	$b \mid c$	
kohlhase-omdoc_F00094	66	0.531	`c<\operatorname{kg} V(a, b)`	$c<\operatorname{kg} V(a, b)$	
kohlhase-omdoc_F00095	68	1.000	`\mathbb{N}`	$\mathbb{N}$	We begin the extended example with the theories in the lower half of
kohlhase-omdoc_F00096	68	1.000	`s(s(s(0)))`	$s(s(s(0)))$	succ. Intuitively, a sort represents an inductively defined set, i.e. it contains
kohlhase-omdoc_F00097	68	1.000	`s`	s	succ. Intuitively, a sort represents an inductively defined set, i.e. it contains
kohlhase-omdoc_F00098	68	1.000	`\leq`	$\leq$	
kohlhase-omdoc_F00099	69	0.887	`\mathrm{TO}^{1}`	TO^{1}	sets and the ordering relations to each other. The obliga
kohlhase-omdoc_F00100	72	1.000	`\sigma(\circ)`	$\sigma(\circ)$	We start off with the theory for *semigroups*. It introduces two symbols, the
kohlhase-omdoc_F00101	72	1.000	`\sigma(\forall x, y \in M . x \circ y \in M)=\forall y, x \in M . x \circ y \in M`	$\sigma(\forall x, y \in M . x \circ y \in M)=\forall y, x \in M . x \circ y \in M$	that M is closed under $\circ$ and that $\circ$ is associative on M. We have a struc-
kohlhase-omdoc_F00102	72	0.835	`\tau x`	τx	<theory xml:id="monoid"> <imports xml:id="sg2mon" from="#semigroup"/>
kohlhase-omdoc_F00103	73	1.000	`\rho`	ρ	Note that all of these axiom inclusions have the same morphism (denoted by
kohlhase-omdoc_F00104	73	0.951	`^{-1}`	$^{-1}$	<theory xml:id="group">
kohlhase-omdoc_F00105	74	1.000	`\sigma`	σ	62 7 A Development Graph for Elementary Algebra
kohlhase-omdoc_F00106	74	1.000	`\tau`	τ	62 7 A Development Graph for Elementary Algebra
kohlhase-omdoc_F00107	74	0.301	`\cdot{ }^{*}`	$\cdot{ }^{*}$	Finally, we extend the whole setup to a theory of rings. Note that we have
kohlhase-omdoc_F00108	74	0.267	`l`	l	</imports>

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_F00109	84	1.000	`\mathcal{N}`	$\mathcal{N}$	earized according to the dependency relation. This is often sufficient, but
kohlhase-omdoc_F00110	84	1.000	`\mathcal{F}`	$\mathcal{F}$	earized according to the dependency relation. This is often sufficient, but
kohlhase-omdoc_F00111	84	1.000	`\mathcal{E}_{\mathcal{N}}`	$\mathcal{E}_{\mathcal{N}}$	earized according to the dependency relation. This is often sufficient, but
kohlhase-omdoc_F00112	84	1.000	`\mathcal{S}`	$\mathcal{S}$	more complex rhetoric/didactic figures are also possible. For instance, when
kohlhase-omdoc_F00113	84	1.000	`\mathcal{E}_{\mathcal{S}}`	$\mathcal{E}_{\mathcal{S}}$	we introduce a new concept, we often first introduce a naive reduced approx- imation $\mathcal{N}$ of the real theory $\mathcal{F}$, only to show an example $\mathcal{E}_{\mathcal{N}}$ of where this
kohlhase-omdoc_F00114	84	1.000	`\mathcal{E}_{\mathcal{F}}`	$\mathcal{E}_{\mathcal{F}}$	example $\mathcal{E}_{\mathcal{S}}$ of why this does not work. Based on the inform
kohlhase-omdoc_F00115	84	1.000	`\mathcal{N}, \mathcal{E}_{\mathcal{N}}, \mathcal{S}, \mathcal{E}_{\mathcal{S}}, \mathcal{F}`	$\mathcal{N}, \mathcal{E}_{\mathcal{N}}, \mathcal{S}, \mathcal{E}_{\mathcal{S}}, \mathcal{F}$	structure with the solid lines and boxes at the bottom of the diagram repre-
kohlhase-omdoc_F00116	84	0.384	`\operatorname{slides} \mathrm{sl}_{i}`	$\operatorname{slides} \mathrm{sl}_{i}$	
kohlhase-omdoc_F00117	84	1.000	`\mathrm{n}_{1}`	n_{1}	
kohlhase-omdoc_F00118	84	0.974	`\mathrm{n}_{2}`	n_{2}	
kohlhase-omdoc_F00119	85	0.986	`\left[\mathrm{MBG}^{+} 03\right]`	$\left[\mathrm{MBG}^{+} 03\right]$	
kohlhase-omdoc_F00120	87	0.810	`9.3{ }^{2}`	$9.3{ }^{2}$	POST http://spass.mpi-sb.mpg.de/webspass/soap HTTP/1.1 Host: http://spass.mpi-sb.mpg.de/webspass/soap
kohlhase-omdoc_F00121	103	1.000	`R`	R	11.5 Sharing Document Parts 95
kohlhase-omdoc_F00122	103	0.978	`\left[\mathrm{SBC}^{+} 00\right]`	$\left[\mathrm{SBC}^{+} 00\right]$	
kohlhase-omdoc_F00123	107	0.910	`\langle\langle Y Y Y Y\rangle\rangle-\langle\langle M M\rangle\rangle-\langle\langle D D\rangle\rangle\mathrm{T}\langle\langle h h\rangle\rangle:\langle\langle m m\rangle\rangle:\langle\langle s s\rangle\rangle`	$\langle\langle Y Y Y Y\rangle\rangle-\langle\langle M M\rangle\rangle-\langle\langle D D\rangle\rangle\mathrm{T}\langle\langle h h\rangle\rangle:\langle\langle m m\rangle\rangle:\langle\langle s s\rangle\rangle$	day, preceded by an optional leading "-" sign to indicate a negative
kohlhase-omdoc_F00124	107	0.781	`\langle\langle Y Y Y Y\rangle\rangle`	$\langle\langle Y Y Y Y\rangle\rangle$	date/time separator and ⟪hh⟫, ⟪mm⟫, ⟪ss⟫ represent hour, minutes, and
kohlhase-omdoc_F00125	107	0.781	`M M\rangle`	$M M\rangle$	date/time separator and ⟪hh⟫, ⟪mm⟫, ⟪ss⟫ represent hour, minutes, and
kohlhase-omdoc_F00126	107	0.781	`\langle\langle D D\rangle`	$\langle\langle D D\rangle$	date/time separator and ⟪hh⟫, ⟪mm⟫, ⟪ss⟫ represent hour, minutes, and
kohlhase-omdoc_F00127	107	0.582	`\langle\langle h h\rangle\rangle,\langle\langle m m \rangle\rangle,\langle\langle s s\rangle\rangle`	$\langle\langle h h\rangle\rangle,\langle\langle m m \rangle\rangle,\langle\langle s s\rangle\rangle$	element has the attributes action and who to specify who did what: The
kohlhase-omdoc_F00128	107	0.260	`\rangle .\langle`	$\rangle .\langle$	and dc is a keyword for the action undertaken. Recommended values in- clude the short forms updated, created, imported, frozen, review-on,
kohlhase-omdoc_F00129	108	1.000	`\widehat{=}`	$\widehat{=}$	element will contain a rights management statement, or reference a ser-
kohlhase-omdoc_F00130	108	0.327	`\mathrm{de} \widehat{=}`	$\mathrm{de} \widehat{=}$	vice providing such information. dc:rights information
kohlhase-omdoc_F00131	114	0.586	`{ }^{+} 03`	${ }^{+} 03$	and the MATHML 2.0 recommendation (second edition) [ABC+03] are part of this specification. We will review OPENMATH objects (top-level element
kohlhase-omdoc_F00132	116	1.000	`\sin (x)`	$\sin (x)$	attribute (see Section 13.4 for further discussion).
kohlhase-omdoc_F00133	116	1.000	`\forall x . \sin (x) \leq \pi`	$\forall x . \sin (x) \leq \pi$	<OMS cd="arith1" name
kohlhase-omdoc_F00134	117	0.999	`x+\pi`	$x+\pi$	In the OPENMATH fragment in Listing 13.3 the expression $x+\pi$ is annotated

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_ F00135	118	1.000	`\sin (\pi)`	$\sin(\pi)$	function sin only allows one argument), and the inner one flags the typo in
kohlhase-omdoc_ F00136	118	1.000	`\pi`	π	<OME>
kohlhase-omdoc_ F00137	121	0.973	`14^{\text {th }}`	14^{th}	MATHML 2.0 that are relevant to OMDOC; for the full story see [ABC+03].
kohlhase-omdoc_ F00138	121	0.630	`2.0\left[\mathrm{ABC}^{+} 03\right]`	$2.0\left[\mathrm{ABC}^{+} 03\right]$	the use of the Content-MATHML fragment described in this section, which
kohlhase-omdoc_ F00139	121	0.997	`\left[\mathrm{ABC}^{+} 03\right]`	$\left[\mathrm{ABC}^{+} 03\right]$	The top-level element of MATHML is the m:math[7] element, see Figure 13.10
kohlhase-omdoc_ F00140	121	0.456	`^{7}`	^{7}	
kohlhase-omdoc_ F00141	124	1.000	`(\mathbb{R} \rightarrow \mathbb{R}) \rightarrow( \mathbb{R} \rightarrow \mathbb{R})`	$(\mathbb{R} \rightarrow \mathbb{R}) \rightarrow(\mathbb{R} \rightarrow \mathbb{R})$	
kohlhase-omdoc_ F00142	126	0.397	`\mathrm{cd}=`	$\mathrm{cd}=$	have to understand the value of the name attribute as a pointer to a defining
kohlhase-omdoc_ F00143	126	0.397	`\left\langle\left\langle c d_{1}\right\rangle\right.`	$\left\langle\left\langle c d_{1}\right\rangle\right.$	have to understand the value of the name attribute as a pointer to a defining
kohlhase-omdoc_ F00144	126	0.175	`\left\langle\left\langle c d_{2}\right\rangle{ }^{\prime \prime}\right.`	$\left\langle\left\langle c d_{2}\right\rangle{ }^{\prime \prime}\right.$	occurrence. In case of symbols, this is the symbol declaration in the con-
kohlhase-omdoc_ F00145	126	0.175	`\left.n a m e_{2}\right\rangle`	$\left.n a m e_{2}\right\rangle$	occurrence. In case of symbols, this is the symbol declaration in the con-
kohlhase-omdoc_ F00146	126	0.973	`\left\langle\left\langle c d_{1}\right\rangle\right\rangle=\left\langle\left\langle c d_{2}\right\rangle\right.`	$\left\langle\left\langle c d_{1}\right\rangle\right\rangle=\left\langle\left\langle c d_{2}\right\rangle\right.$	name="⟨name1⟩"/> is equal to <OMS cd="⟨cd2⟩" name="⟨name2⟩"/>, iff
kohlhase-omdoc_ F00147	126	0.973	`\left\langle\left\langle n a m e_{1}\right\rangle\right\rangle=\left\langle\left\langle n a m e_{2}\right\rangle\right.`	$\left\langle\left\langle n a m e_{1}\right\rangle\right\rangle=\left\langle\left\langle n a m e_{2}\right\rangle\right.$	name="⟨name1⟩"/> is equal to <OMS cd="⟨cd2⟩" name="⟨name2⟩"/>, iff
kohlhase-omdoc_ F00148	126	0.923	`\alpha`	α	is not bound, then the scope of the variable cannot be reliably defined; so
kohlhase-omdoc_ F00149	129	0.990	`\mathrm{CMP}^{2}`	CMP^{2}	are written in. Conforming with the XML recommendation, we use the ISO
kohlhase-omdoc_ F00150	130	0.942	`\mathrm{FMP}^{4}`	FMP^{4}	used in formalizing the content. All members of the group have to formalize
kohlhase-omdoc_ F00151	131	0.287	`V, F. Im (F) = V`	$V, F. Im (F) = V$	
kohlhase-omdoc_ F00152	131	0.404	`A</ \mathrm{FMP}>`	$A</ \mathrm{FMP}>$	
kohlhase-omdoc_ F00153	134	0.562	`0 \in \mathbb{N}`	$0 \in \mathbb{N}$	<OMS cd="nat" name="zero"/>
kohlhase-omdoc_ F00154	137	0.995	`\underline{\mathrm{ul}}`	$\underline{\mathrm{ul}}$	
kohlhase-omdoc_ F00155	140	1.000	`s(n)`	$s(n)$	
kohlhase-omdoc_ F00156	140	0.997	`0, s`	$0, s$	
kohlhase-omdoc_ F00157	141	1.000	`\mathcal{T}`	$\mathcal{T}$	
kohlhase-omdoc_ F00158	141	1.000	`s=`	$s=$	
kohlhase-omdoc_ F00159	141	0.623	`\mathbf{A}`	$\mathbf{A}$	theories. As consistency can easily be lost by adding axioms, mathematicians
kohlhase-omdoc_ F00160	141	1.000	`s=\mathbf{A}`	$s=\mathbf{A}$	Sometimes, we can determine that an axiom does not destroy consistency

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_ F00161	141	1.000	`\mathcal{A}`	$\mathcal{A}$	endangering consistency — to theories.
kohlhase-omdoc_ F00162	141	0.554	`1:=s(0)`	$1:=s(0)$	The exponential function $e^{\cdot}$ — The **exponential function** $e^{\cdot}$ is the solution to the differential equation $\partial f = f$ [where $f(0)=1$]. — implicit The addition function + — **Addition** on the natural numbers is defined by the equations $x+0=x$ and $x+s(y)=s(x+y)$. — recursive **Fig. 15.2.** Some common definitions
kohlhase-omdoc_ F00163	141	0.554	`e^{\cdot}`	$e^{\cdot}$	The exponential function $e^{\cdot}$ — The **exponential function** $e^{\cdot}$ is the solution to the differential equation $\partial f = f$ [where $f(0)=1$]. — implicit The addition function + — **Addition** on the natural numbers is defined by the equations $x+0=x$ and $x+s(y)=s(x+y)$. — recursive **Fig. 15.2.** Some common definitions
kohlhase-omdoc_ F00164	141	0.554	`\partial f=f[`	$\partial f=f[$	The exponential function $e^{\cdot}$ — The **exponential function** $e^{\cdot}$ is the solution to the differential equation $\partial f = f$ [where $f(0)=1$]. — implicit The addition function + — **Addition** on the natural numbers is defined by the equations $x+0=x$ and $x+s(y)=s(x+y)$. — recursive **Fig. 15.2.** Some common definitions
kohlhase-omdoc_ F00165	141	0.554	`f(0)=1]`	$f(0)=1]$	The exponential function $e^{\cdot}$ — The **exponential function** $e^{\cdot}$ is the solution to the differential equation $\partial f = f$ [where $f(0)=1$]. — implicit The addition function + — **Addition** on the natural numbers is defined by the equations $x+0=x$ and $x+s(y)=s(x+y)$. — recursive **Fig. 15.2.** Some common definitions
kohlhase-omdoc_ F00166	141	0.554	`x+0=x`	$x+0=x$	The exponential function $e^{\cdot}$ — The **exponential function** $e^{\cdot}$ is the solution to the differential equation $\partial f = f$ [where $f(0)=1$]. — implicit The addition function + — **Addition** on the natural numbers is defined by the equations $x+0=x$ and $x+s(y)=s(x+y)$. — recursive **Fig. 15.2.** Some common definitions
kohlhase-omdoc_ F00167	141	0.554	`x+s(y)=s(x+y)`	$x+s(y)=s(x+y)$	The exponential function $e^{\cdot}$ — The **exponential function** $e^{\cdot}$ is the solution to the differential equation $\partial f = f$ [where $f(0)=1$]. — implicit The addition function + — **Addition** on the natural numbers is defined by the equations $x+0=x$ and $x+s(y)=s(x+y)$. — recursive **Fig. 15.2.** Some common definitions
kohlhase-omdoc_ F00168	143	0.993	`1.1\left[\mathrm{BPSM}^{+} 04\right]`	$1.1\left[\mathrm{BPSM}^{+} 04\right]$	The `symbol` element has a required attribute `name` whose value uniquely
kohlhase-omdoc_ F00169	144	1.000	`e: t`	$e: t$	
kohlhase-omdoc_ F00170	145	0.998	`t`	t	
kohlhase-omdoc_ F00171	145	1.000	`\sum_{i=1}^{n} a_{i} x^{i}`	$\sum_{i=1}^{n} a_{i} x^{i}$	15.2 Theory-Constitutive Statements in OMDoc 139
kohlhase-omdoc_ F00172	145	1.000	`a_{i}`	a_{i}	15.2 Theory-Constitutive Statements in OMDoc 139

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_F00173	145	0.999	`\forall x_{1}, \ldots, x_{n} . e \in t`	$\forall x_{1}, \ldots, x_{n} . e \in t$	 URI reference that points to an assertion or axiom element that asserts
kohlhase-omdoc_F00174	145	0.999	`x_{1}, \ldots, x_{n}`	$x_{1}, \ldots, x_{n}$	 URI reference that points to an assertion or axiom element that asserts
kohlhase-omdoc_F00175	148	0.397	`\rangle\rangle *`	$\rangle\rangle *$	
kohlhase-omdoc_F00176	149	0.944	`x+x`	$x+x$	
kohlhase-omdoc_F00177	150	1.000	`\mathcal{C}_{i}`	$\mathcal{C}_{i}$	

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Element	Attributes		D	Content
	Required	Optional	C	
assertion		xml:id, for, type, theory, class, style, status, just-by	+	CMP*, FMP*
type	system	xml:id, for, just-by, theory, class, style	−	CMP*, $\langle\!\langle mobj\rangle\!\rangle$, $\langle\!\langle mobj\rangle\!\rangle$
example	for	xml:id, type, assertion, theory, class, style	+	CMP* | $\langle\!\langle mobj\rangle\!\rangle$*
alternative	for, theory, entailed-by, entails,	xml:id, type, theory, class, style	+	CMP*, (FMP* | requation* | $\langle\!\langle mobj\rangle\!\rangle$)

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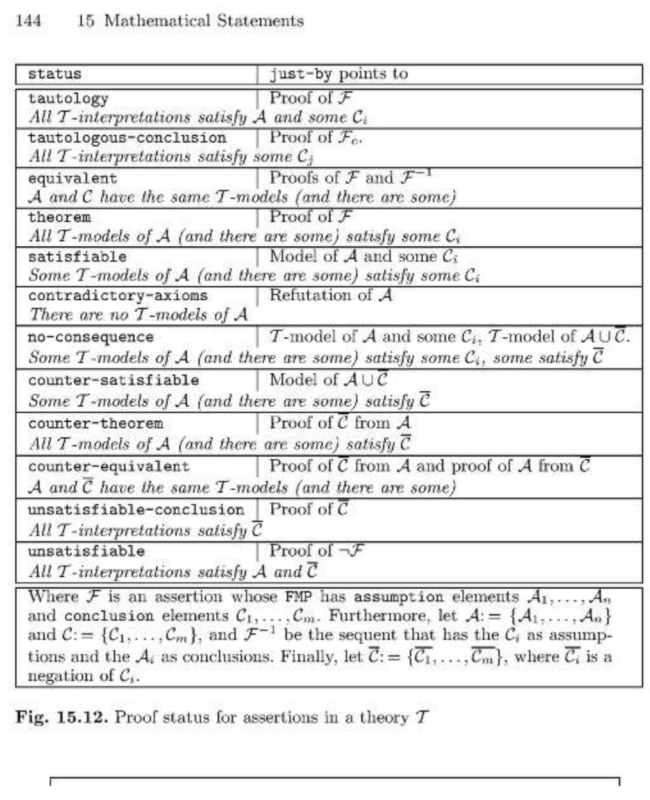
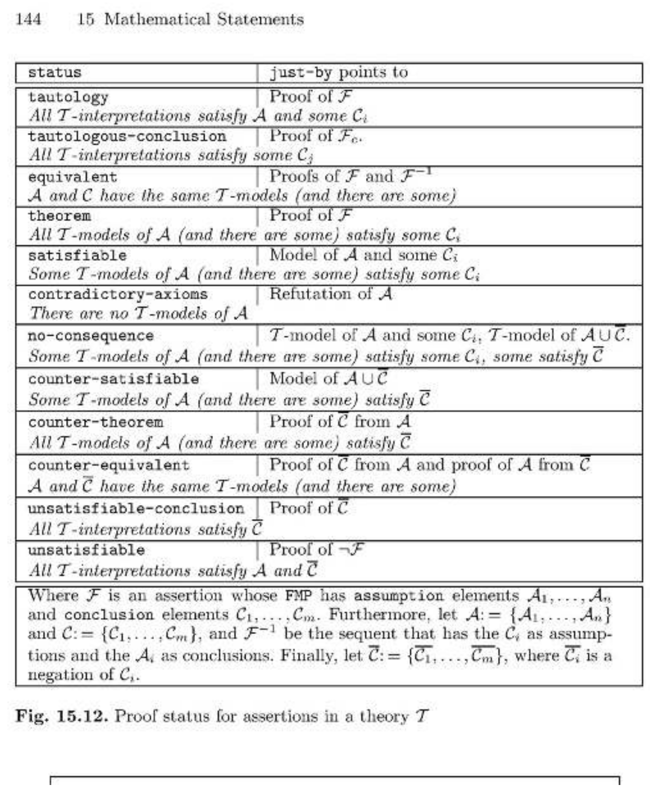
status	just-by points to
tautology *All $\mathcal{T}$-interpretations satisfy $\mathcal{A}$ and some $\mathcal{C}_i$*	Proof of $\mathcal{F}$
tautologous-conclusion *All $\mathcal{T}$-interpretations satisfy some $\mathcal{C}_j$*	Proof of $\mathcal{F}_c$.
equivalent *$\mathcal{A}$ and $\mathcal{C}$ have the same $\mathcal{T}$-models (and there are some)*	Proofs of $\mathcal{F}$ and $\mathcal{F}^{-1}$
theorem *All $\mathcal{T}$-models of $\mathcal{A}$ (and there are some) satisfy some $\mathcal{C}_i$*	Proof of $\mathcal{F}$
satisfiable *Some $\mathcal{T}$-models of $\mathcal{A}$ (and there are some) satisfy some $\mathcal{C}_i$*	Model of $\mathcal{A}$ and some $\mathcal{C}_i$
contradictory-axioms *There are no $\mathcal{T}$-models of $\mathcal{A}$*	Refutation of $\mathcal{A}$
no-consequence *Some $\mathcal{T}$-models of $\mathcal{A}$ (and there are some) satisfy some $\mathcal{C}_i$, some satisfy $\overline{\mathcal{C}}$*	$\mathcal{T}$-model of $\mathcal{A}$ and some $\mathcal{C}_i$, $\mathcal{T}$-model of $\mathcal{A}\cup\overline{\mathcal{C}}$.
counter-satisfiable *Some $\mathcal{T}$-models of $\mathcal{A}$ (and there are some) satisfy $\overline{\mathcal{C}}$*	Model of $\mathcal{A}\cup\overline{\mathcal{C}}$
counter-theorem *All $\mathcal{T}$-models of $\mathcal{A}$ (and there are some) satisfy $\overline{\mathcal{C}}$*	Proof of $\overline{\mathcal{C}}$ from $\mathcal{A}$
counter-equivalent *$\mathcal{A}$ and $\overline{\mathcal{C}}$ have the same $\mathcal{T}$-models (and there are some)*	Proof of $\overline{\mathcal{C}}$ from $\mathcal{A}$ and proof of $\mathcal{A}$ from $\overline{\mathcal{C}}$
unsatisfiable-conclusion *All $\mathcal{T}$-interpretations satisfy $\overline{\mathcal{C}}$*	Proof of $\overline{\mathcal{C}}$
unsatisfiable *All $\mathcal{T}$-interpretations satisfy $\mathcal{A}$ and $\overline{\mathcal{C}}$*	Proof of $\neg\mathcal{F}$

Where $\mathcal{F}$ is an assertion whose FMP has assumption elements $\mathcal{A}_1,\ldots,\mathcal{A}_n$ and conclusion elements $\mathcal{C}_1,\ldots,\mathcal{C}_m$. Furthermore, let $\mathcal{A}:=\{\mathcal{A}_1,\ldots,\mathcal{A}_n\}$ and $\mathcal{C}:=\{\mathcal{C}_1,\ldots,\mathcal{C}_m\}$, and $\mathcal{F}^{-1}$ be the sequent that has the $\mathcal{C}_i$ as assumptions and the $\mathcal{A}_i$ as conclusions. Finally, let $\overline{\mathcal{C}}:=\{\overline{\mathcal{C}_1},\ldots,\overline{\mathcal{C}_m}\}$, where $\overline{\mathcal{C}_i}$ is a negation of $\mathcal{C}_i$.

Fig. 15.12. Proof status for assertions in a theory $\mathcal{T}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_F00178	150	■ 1.000	\mathcal{F}_{c}	$\mathcal{F}_c$	
kohlhase-omdoc_F00179	150	■ 1.000	\mathcal{C}_{j}	$\mathcal{C}_j$	

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status	just-by points to
tautology *All $\mathcal{T}$-interpretations satisfy $\mathcal{A}$ and some $\mathcal{C}_i$*	Proof of $\mathcal{F}$
tautologous-conclusion *All $\mathcal{T}$-interpretations satisfy some $\mathcal{C}_j$*	Proof of $\mathcal{F}_c$.
equivalent *$\mathcal{A}$ and $\mathcal{C}$ have the same $\mathcal{T}$-models (and there are some)*	Proofs of $\mathcal{F}$ and $\mathcal{F}^{-1}$
theorem *All $\mathcal{T}$-models of $\mathcal{A}$ (and there are some) satisfy some $\mathcal{C}_i$*	Proof of $\mathcal{F}$
satisfiable *Some $\mathcal{T}$-models of $\mathcal{A}$ (and there are some) satisfy some $\mathcal{C}_i$*	Model of $\mathcal{A}$ and some $\mathcal{C}_i$
contradictory-axioms *There are no $\mathcal{T}$-models of $\mathcal{A}$*	Refutation of $\mathcal{A}$
no-consequence *Some $\mathcal{T}$-models of $\mathcal{A}$ (and there are some) satisfy some $\mathcal{C}_i$, some satisfy $\overline{\mathcal{C}}$*	$\mathcal{T}$-model of $\mathcal{A}$ and some $\mathcal{C}_i$, $\mathcal{T}$-model of $\mathcal{A} \cup \overline{\mathcal{C}}$.
counter-satisfiable *Some $\mathcal{T}$-models of $\mathcal{A}$ (and there are some) satisfy $\overline{\mathcal{C}}$*	Model of $\mathcal{A} \cup \overline{\mathcal{C}}$
counter-theorem *All $\mathcal{T}$-models of $\mathcal{A}$ (and there are some) satisfy $\overline{\mathcal{C}}$*	Proof of $\overline{\mathcal{C}}$ from $\mathcal{A}$
counter-equivalent *$\mathcal{A}$ and $\overline{\mathcal{C}}$ have the same $\mathcal{T}$-models (and there are some)*	Proof of $\overline{\mathcal{C}}$ from $\mathcal{A}$ and proof of $\mathcal{A}$ from $\overline{\mathcal{C}}$
unsatisfiable-conclusion *All $\mathcal{T}$-interpretations satisfy $\overline{\mathcal{C}}$*	Proof of $\overline{\mathcal{C}}$
unsatisfiable *All $\mathcal{T}$-interpretations satisfy $\mathcal{A}$ and $\overline{\mathcal{C}}$*	Proof of $\neg\mathcal{F}$

Where $\mathcal{F}$ is an assertion whose FMP has assumption elements $\mathcal{A}_1, \ldots, \mathcal{A}_n$ and conclusion elements $\mathcal{C}_1, \ldots, \mathcal{C}_m$. Furthermore, let $\mathcal{A} := \{\mathcal{A}_1, \ldots, \mathcal{A}_n\}$ and $\mathcal{C} := \{\mathcal{C}_1, \ldots, \mathcal{C}_m\}$, and $\mathcal{F}^{-1}$ be the sequent that has the $\mathcal{C}_i$ as assumptions and the $\mathcal{A}_i$ as conclusions. Finally, let $\overline{\mathcal{C}} := \{\overline{\mathcal{C}_1}, \ldots, \overline{\mathcal{C}_m}\}$, where $\overline{\mathcal{C}_i}$ is a negation of $\mathcal{C}_i$.

Fig. 15.12. Proof status for assertions in a theory $\mathcal{T}$

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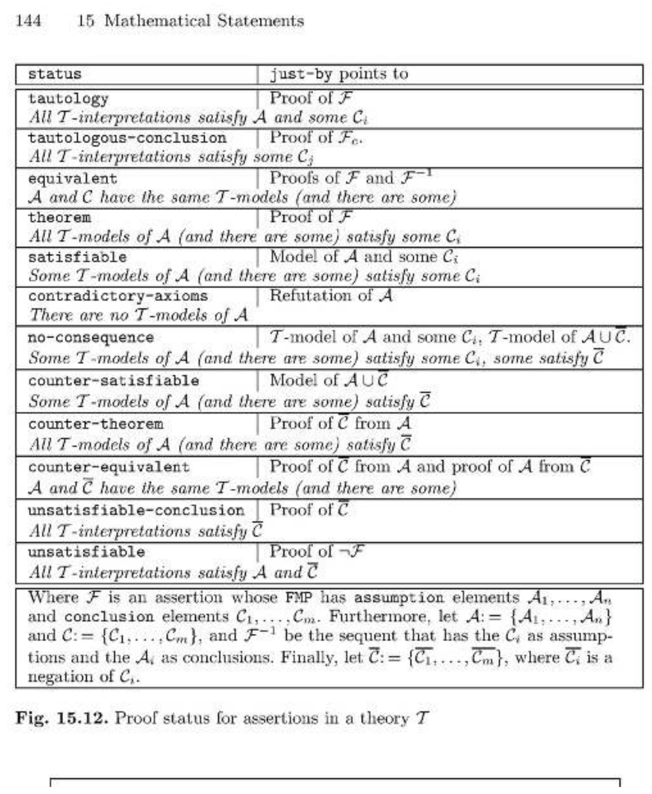
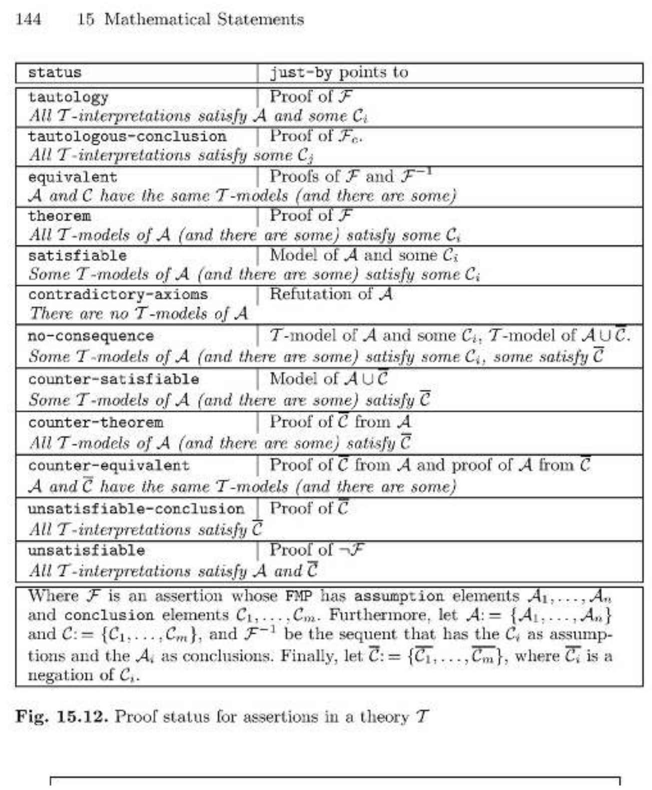
status	just-by points to
tautology *All $\mathcal{T}$-interpretations satisfy $\mathcal{A}$ and some $\mathcal{C}_i$*	Proof of $\mathcal{F}$
tautologous-conclusion *All $\mathcal{T}$-interpretations satisfy some $\mathcal{C}_j$*	Proof of $\mathcal{F}_c$.
equivalent *$\mathcal{A}$ and $\mathcal{C}$ have the same $\mathcal{T}$-models (and there are some)*	Proofs of $\mathcal{F}$ and $\mathcal{F}^{-1}$
theorem *All $\mathcal{T}$-models of $\mathcal{A}$ (and there are some) satisfy some $\mathcal{C}_i$*	Proof of $\mathcal{F}$
satisfiable *Some $\mathcal{T}$-models of $\mathcal{A}$ (and there are some) satisfy some $\mathcal{C}_i$*	Model of $\mathcal{A}$ and some $\mathcal{C}_i$
contradictory-axioms *There are no $\mathcal{T}$-models of $\mathcal{A}$*	Refutation of $\mathcal{A}$
no-consequence *Some $\mathcal{T}$-models of $\mathcal{A}$ (and there are some) satisfy some $\mathcal{C}_i$, some satisfy $\overline{\mathcal{C}}$*	$\mathcal{T}$-model of $\mathcal{A}$ and some $\mathcal{C}_i$, $\mathcal{T}$-model of $\mathcal{A} \cup \overline{\mathcal{C}}$.
counter-satisfiable *Some $\mathcal{T}$-models of $\mathcal{A}$ (and there are some) satisfy $\overline{\mathcal{C}}$*	Model of $\mathcal{A} \cup \overline{\mathcal{C}}$
counter-theorem *All $\mathcal{T}$-models of $\mathcal{A}$ (and there are some) satisfy $\overline{\mathcal{C}}$*	Proof of $\overline{\mathcal{C}}$ from $\mathcal{A}$
counter-equivalent *$\mathcal{A}$ and $\overline{\mathcal{C}}$ have the same $\mathcal{T}$-models (and there are some)*	Proof of $\overline{\mathcal{C}}$ from $\mathcal{A}$ and proof of $\mathcal{A}$ from $\overline{\mathcal{C}}$
unsatisfiable-conclusion *All $\mathcal{T}$-interpretations satisfy $\overline{\mathcal{C}}$*	Proof of $\overline{\mathcal{C}}$
unsatisfiable *All $\mathcal{T}$-interpretations satisfy $\mathcal{A}$ and $\overline{\mathcal{C}}$*	Proof of $\neg\mathcal{F}$

Where $\mathcal{F}$ is an assertion whose FMP has assumption elements $\mathcal{A}_1, \ldots, \mathcal{A}_n$ and conclusion elements $\mathcal{C}_1, \ldots, \mathcal{C}_m$. Furthermore, let $\mathcal{A} := \{\mathcal{A}_1, \ldots, \mathcal{A}_n\}$ and $\mathcal{C} := \{\mathcal{C}_1, \ldots, \mathcal{C}_m\}$, and $\mathcal{F}^{-1}$ be the sequent that has the $\mathcal{C}_i$ as assumptions and the $\mathcal{A}_i$ as conclusions. Finally, let $\overline{\mathcal{C}} := \{\overline{\mathcal{C}_1}, \ldots, \overline{\mathcal{C}_m}\}$, where $\overline{\mathcal{C}_i}$ is a negation of $\mathcal{C}_i$.

Fig. 15.12. Proof status for assertions in a theory $\mathcal{T}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_ F00180	150	■ 1.000	\mathcal{F}^{-1}	$\mathcal{F}^{-1}$	
kohlhase-omdoc_ F00181	150	■ 1.000	\mathcal{C}	$\mathcal{C}$	

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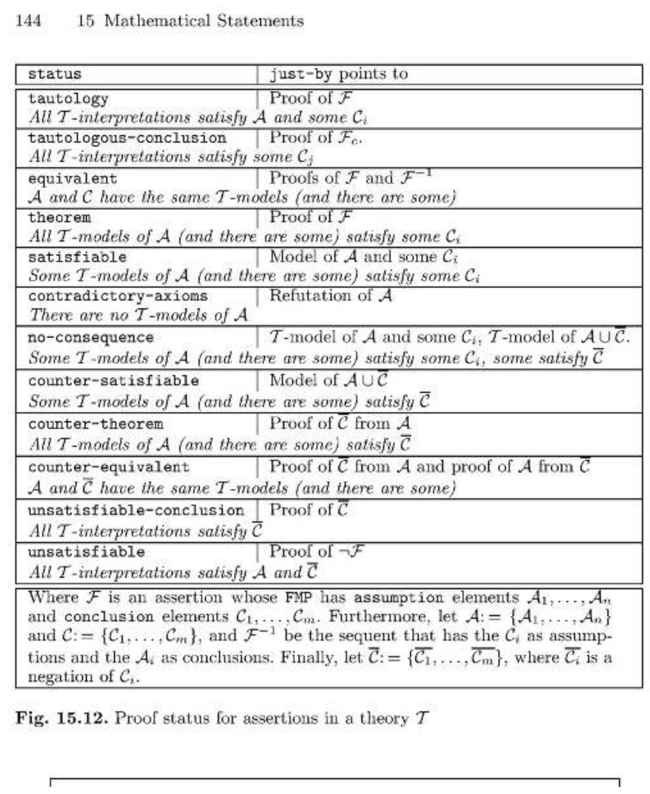
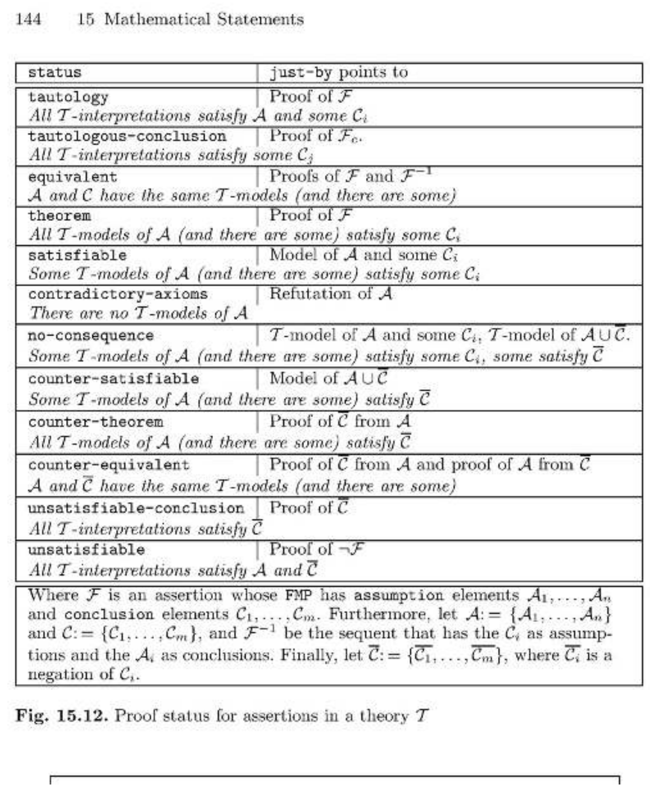
status	just-by points to
tautology *All $\mathcal{T}$-interpretations satisfy $\mathcal{A}$ and some $\mathcal{C}_i$*	Proof of $\mathcal{F}$
tautologous-conclusion *All $\mathcal{T}$-interpretations satisfy some $\mathcal{C}_j$*	Proof of $\mathcal{F}_c$.
equivalent *$\mathcal{A}$ and $\mathcal{C}$ have the same $\mathcal{T}$-models (and there are some)*	Proofs of $\mathcal{F}$ and $\mathcal{F}^{-1}$
theorem *All $\mathcal{T}$-models of $\mathcal{A}$ (and there are some) satisfy some $\mathcal{C}_i$*	Proof of $\mathcal{F}$
satisfiable *Some $\mathcal{T}$-models of $\mathcal{A}$ (and there are some) satisfy some $\mathcal{C}_i$*	Model of $\mathcal{A}$ and some $\mathcal{C}_i$
contradictory-axioms *There are no $\mathcal{T}$-models of $\mathcal{A}$*	Refutation of $\mathcal{A}$
no-consequence *Some $\mathcal{T}$-models of $\mathcal{A}$ (and there are some) satisfy some $\mathcal{C}_i$, some satisfy $\overline{\mathcal{C}}$*	$\mathcal{T}$-model of $\mathcal{A}$ and some $\mathcal{C}_i$, $\mathcal{T}$-model of $\mathcal{A}\cup\overline{\mathcal{C}}$.
counter-satisfiable *Some $\mathcal{T}$-models of $\mathcal{A}$ (and there are some) satisfy $\overline{\mathcal{C}}$*	Model of $\mathcal{A}\cup\overline{\mathcal{C}}$
counter-theorem *All $\mathcal{T}$-models of $\mathcal{A}$ (and there are some) satisfy $\overline{\mathcal{C}}$*	Proof of $\overline{\mathcal{C}}$ from $\mathcal{A}$
counter-equivalent *$\mathcal{A}$ and $\overline{\mathcal{C}}$ have the same $\mathcal{T}$-models (and there are some)*	Proof of $\overline{\mathcal{C}}$ from $\mathcal{A}$ and proof of $\mathcal{A}$ from $\overline{\mathcal{C}}$
unsatisfiable-conclusion *All $\mathcal{T}$-interpretations satisfy $\overline{\mathcal{C}}$*	Proof of $\overline{\mathcal{C}}$
unsatisfiable *All $\mathcal{T}$-interpretations satisfy $\mathcal{A}$ and $\overline{\mathcal{C}}$*	Proof of $\neg\mathcal{F}$

Where $\mathcal{F}$ is an assertion whose FMP has assumption elements $\mathcal{A}_1,\ldots,\mathcal{A}_n$ and conclusion elements $\mathcal{C}_1,\ldots,\mathcal{C}_m$. Furthermore, let $\mathcal{A}:=\{\mathcal{A}_1,\ldots,\mathcal{A}_n\}$ and $\mathcal{C}:=\{\mathcal{C}_1,\ldots,\mathcal{C}_m\}$, and $\mathcal{F}^{-1}$ be the sequent that has the $\mathcal{C}_i$ as assumptions and the $\mathcal{A}_i$ as conclusions. Finally, let $\overline{\mathcal{C}}:=\{\overline{\mathcal{C}_1},\ldots,\overline{\mathcal{C}_m}\}$, where $\overline{\mathcal{C}_i}$ is a negation of $\mathcal{C}_i$.

Fig. 15.12. Proof status for assertions in a theory $\mathcal{T}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_F00182	150	■ 1.000	`\mathcal{C}_{i}, \mathcal{T}`	$\mathcal{C}_{i}, \mathcal{T}$	
kohlhase-omdoc_F00183	150	■ 1.000	`\mathcal{A} \cup \overline{\mathcal{C}}`	$\mathcal{A} \cup \overline{\mathcal{C}}$	

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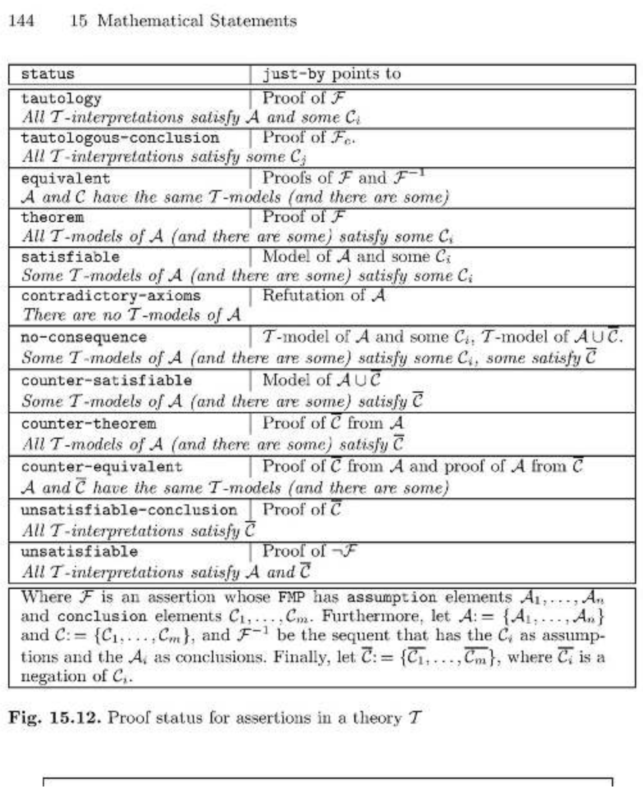
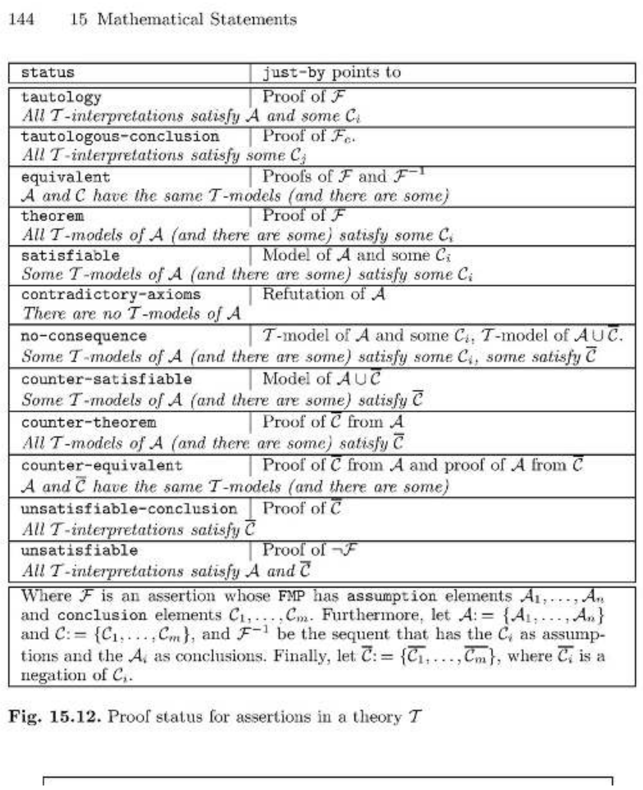
status	just-by points to
tautology	Proof of $\mathcal{F}$
All $\mathcal{T}$-interpretations satisfy $\mathcal{A}$ and some $\mathcal{C}_i$	
tautologous-conclusion	Proof of $\mathcal{F}_c$.
All $\mathcal{T}$-interpretations satisfy some $\mathcal{C}_j$	
equivalent	Proofs of $\mathcal{F}$ and $\mathcal{F}^{-1}$
$\mathcal{A}$ and $\mathcal{C}$ have the same $\mathcal{T}$-models (and there are some)	
theorem	Proof of $\mathcal{F}$
All $\mathcal{T}$-models of $\mathcal{A}$ (and there are some) satisfy some $\mathcal{C}_i$	
satisfiable	Model of $\mathcal{A}$ and some $\mathcal{C}_i$
Some $\mathcal{T}$-models of $\mathcal{A}$ (and there are some) satisfy some $\mathcal{C}_i$	
contradictory-axioms	Refutation of $\mathcal{A}$
There are no $\mathcal{T}$-models of $\mathcal{A}$	
no-consequence	$\mathcal{T}$-model of $\mathcal{A}$ and some $\mathcal{C}_i$, $\mathcal{T}$-model of $\mathcal{A} \cup \overline{\mathcal{C}}$.
Some $\mathcal{T}$-models of $\mathcal{A}$ (and there are some) satisfy some $\mathcal{C}_i$, some satisfy $\overline{\mathcal{C}}$	
counter-satisfiable	Model of $\mathcal{A} \cup \overline{\mathcal{C}}$
Some $\mathcal{T}$-models of $\mathcal{A}$ (and there are some) satisfy $\overline{\mathcal{C}}$	
counter-theorem	Proof of $\overline{\mathcal{C}}$ from $\mathcal{A}$
All $\mathcal{T}$-models of $\mathcal{A}$ (and there are some) satisfy $\overline{\mathcal{C}}$	
counter-equivalent	Proof of $\overline{\mathcal{C}}$ from $\mathcal{A}$ and proof of $\mathcal{A}$ from $\overline{\mathcal{C}}$
$\mathcal{A}$ and $\overline{\mathcal{C}}$ have the same $\mathcal{T}$-models (and there are some)	
unsatisfiable-conclusion	Proof of $\overline{\mathcal{C}}$
All $\mathcal{T}$-interpretations satisfy $\overline{\mathcal{C}}$	
unsatisfiable	Proof of $\neg\mathcal{F}$
All $\mathcal{T}$-interpretations satisfy $\mathcal{A}$ and $\overline{\mathcal{C}}$	

Where $\mathcal{F}$ is an assertion whose FMP has assumption elements $\mathcal{A}_1, \ldots, \mathcal{A}_n$ and conclusion elements $\mathcal{C}_1, \ldots, \mathcal{C}_m$. Furthermore, let $\mathcal{A} := \{\mathcal{A}_1, \ldots, \mathcal{A}_n\}$ and $\mathcal{C} := \{\mathcal{C}_1, \ldots, \mathcal{C}_m\}$, and $\mathcal{F}^{-1}$ be the sequent that has the $\mathcal{C}_i$ as assumptions and the $\mathcal{A}_i$ as conclusions. Finally, let $\overline{\mathcal{C}} := \{\overline{\mathcal{C}_1}, \ldots, \overline{\mathcal{C}_m}\}$, where $\overline{\mathcal{C}_i}$ is a negation of $\mathcal{C}_i$.

Fig. 15.12. Proof status for assertions in a theory $\mathcal{T}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_F00184	150	■ 1.000	`\overline{\mathcal{C}}`	$\overline{\mathcal{C}}$	
kohlhase-omdoc_F00185	150	■ 1.000	`\neg \mathcal{F}`	$\neg \mathcal{F}$	

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status	just-by points to
`tautology`	Proof of $\mathcal{F}$
All $\mathcal{T}$-interpretations satisfy $\mathcal{A}$ and some $\mathcal{C}_i$	
`tautologous-conclusion`	Proof of $\mathcal{F}_c$.
All $\mathcal{T}$-interpretations satisfy some $\mathcal{C}_j$	
`equivalent`	Proofs of $\mathcal{F}$ and $\mathcal{F}^{-1}$
$\mathcal{A}$ and $\mathcal{C}$ have the same $\mathcal{T}$-models (and there are some)	
`theorem`	Proof of $\mathcal{F}$
All $\mathcal{T}$-models of $\mathcal{A}$ (and there are some) satisfy some $\mathcal{C}_i$	
`satisfiable`	Model of $\mathcal{A}$ and some $\mathcal{C}_i$
Some $\mathcal{T}$-models of $\mathcal{A}$ (and there are some) satisfy some $\mathcal{C}_i$	
`contradictory-axioms`	Refutation of $\mathcal{A}$
There are no $\mathcal{T}$-models of $\mathcal{A}$	
`no-consequence`	$\mathcal{T}$-model of $\mathcal{A}$ and some $\mathcal{C}_i$, $\mathcal{T}$-model of $\mathcal{A} \cup \overline{\mathcal{C}}$.
Some $\mathcal{T}$-models of $\mathcal{A}$ (and there are some) satisfy some $\mathcal{C}_i$, some satisfy $\overline{\mathcal{C}}$	
`counter-satisfiable`	Model of $\mathcal{A} \cup \overline{\mathcal{C}}$
Some $\mathcal{T}$-models of $\mathcal{A}$ (and there are some) satisfy $\overline{\mathcal{C}}$	
`counter-theorem`	Proof of $\overline{\mathcal{C}}$ from $\mathcal{A}$
All $\mathcal{T}$-models of $\mathcal{A}$ (and there are some) satisfy $\overline{\mathcal{C}}$	
`counter-equivalent`	Proof of $\overline{\mathcal{C}}$ from $\mathcal{A}$ and proof of $\mathcal{A}$ from $\overline{\mathcal{C}}$
$\mathcal{A}$ and $\overline{\mathcal{C}}$ have the same $\mathcal{T}$-models (and there are some)	
`unsatisfiable-conclusion`	Proof of $\overline{\mathcal{C}}$
All $\mathcal{T}$-interpretations satisfy $\overline{\mathcal{C}}$	
`unsatisfiable`	Proof of $\neg\mathcal{F}$
All $\mathcal{T}$-interpretations satisfy $\mathcal{A}$ and $\overline{\mathcal{C}}$	

Where $\mathcal{F}$ is an assertion whose `FMP` has `assumption` elements $\mathcal{A}_1, \ldots, \mathcal{A}_n$ and `conclusion` elements $\mathcal{C}_1, \ldots, \mathcal{C}_m$. Furthermore, let $\mathcal{A} := \{\mathcal{A}_1, \ldots, \mathcal{A}_n\}$ and $\mathcal{C} := \{\mathcal{C}_1, \ldots, \mathcal{C}_m\}$, and $\mathcal{F}^{-1}$ be the sequent that has the $\mathcal{C}_i$ as assumptions and the $\mathcal{A}_i$ as conclusions. Finally, let $\overline{\mathcal{C}} := \{\overline{\mathcal{C}_1}, \ldots, \overline{\mathcal{C}_m}\}$, where $\overline{\mathcal{C}_i}$ is a negation of $\mathcal{C}_i$.

Fig. 15.12. Proof status for assertions in a theory $\mathcal{T}$

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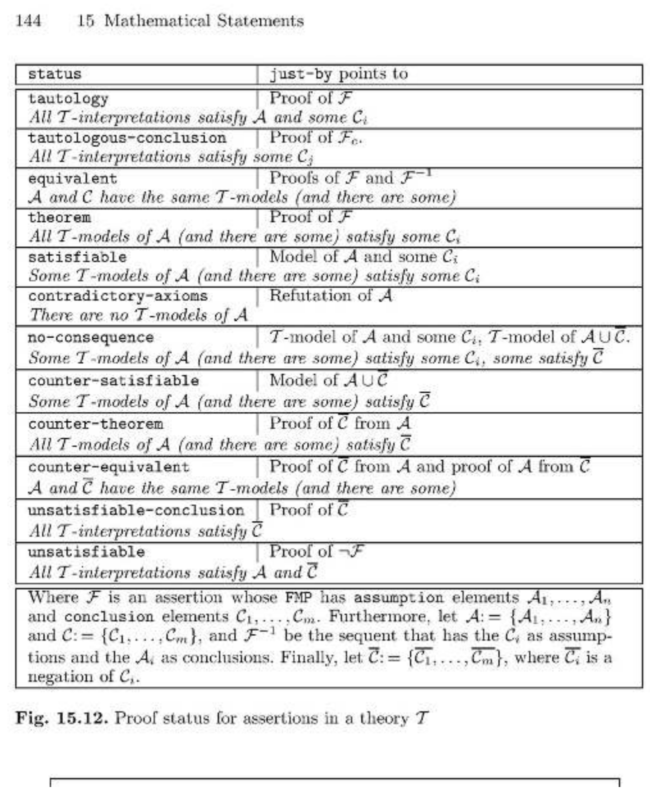
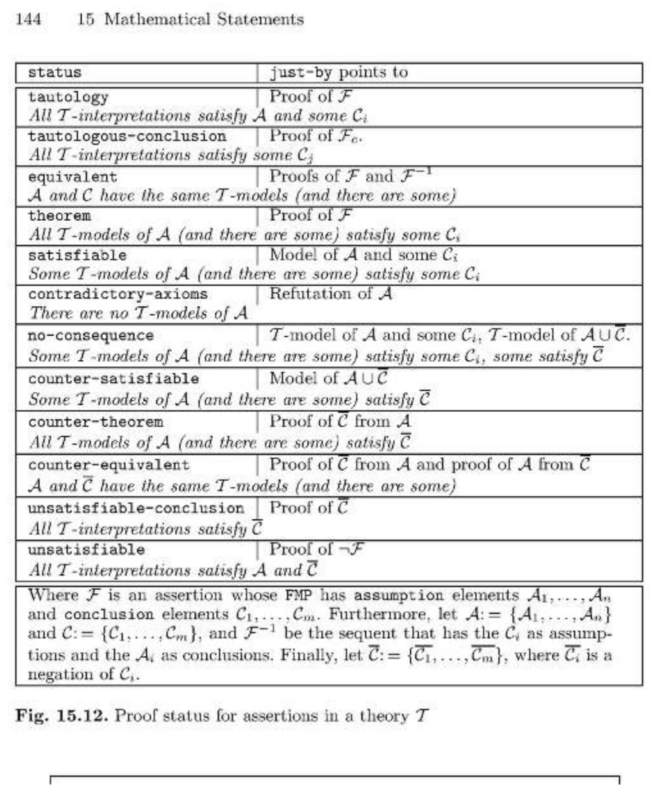
Fig. 15.12. Proof status for assertions in a theory $\mathcal{T}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_F00186	150	■ 1.000	`\mathcal{A}_{1}, \ldots, \mathcal{A}_{n}`	$\mathcal{A}_{1}, \ldots, \mathcal{A}_{n}$	
kohlhase-omdoc_F00187	150	■ 1.000	`\mathcal{C}_{1}, \ldots, \mathcal{C}_{m}`	$\mathcal{C}_{1}, \ldots, \mathcal{C}_{m}$	

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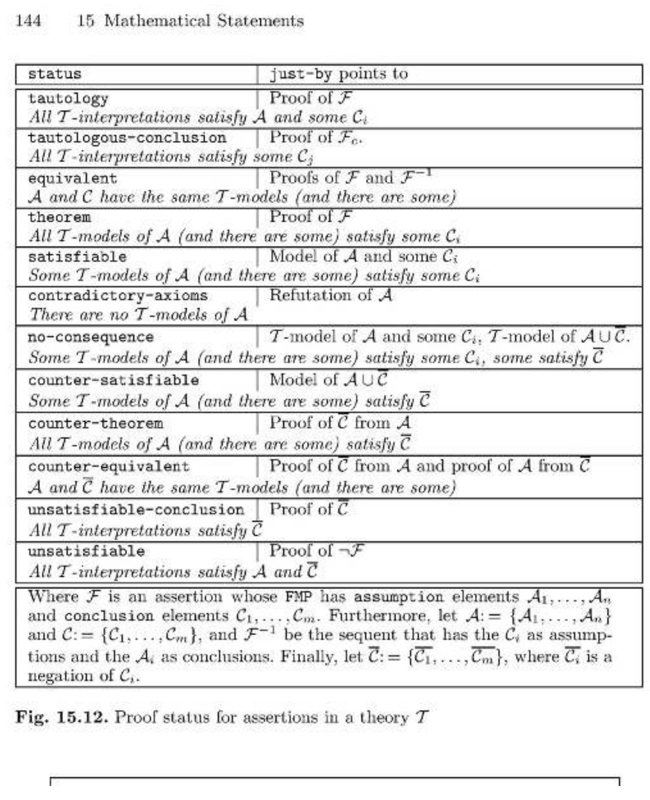
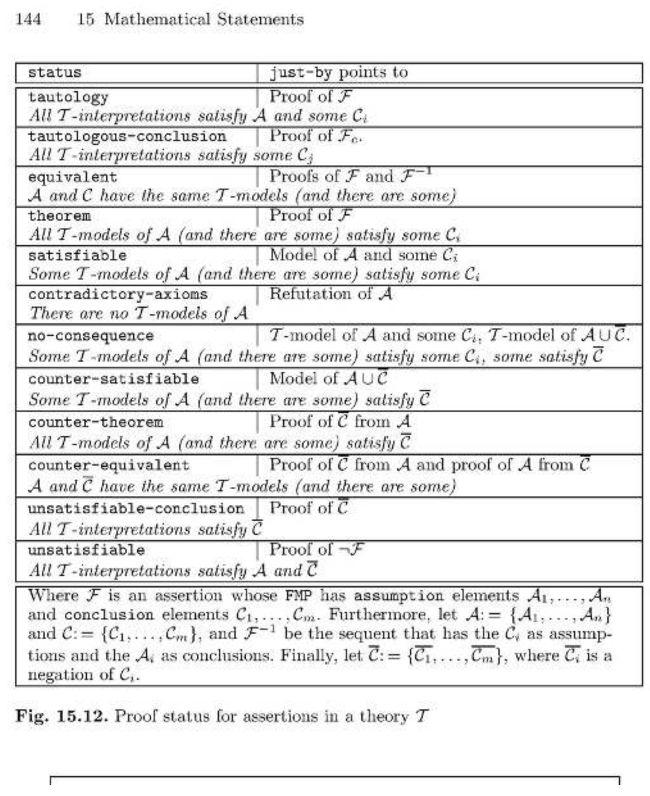
status	just-by points to
tautology *All $\mathcal{T}$-interpretations satisfy $\mathcal{A}$ and some $\mathcal{C}_i$*	Proof of $\mathcal{F}$
tautologous-conclusion *All $\mathcal{T}$-interpretations satisfy some $\mathcal{C}_j$*	Proof of $\mathcal{F}_c$.
equivalent *$\mathcal{A}$ and $\mathcal{C}$ have the same $\mathcal{T}$-models (and there are some)*	Proofs of $\mathcal{F}$ and $\mathcal{F}^{-1}$
theorem *All $\mathcal{T}$-models of $\mathcal{A}$ (and there are some) satisfy some $\mathcal{C}_i$*	Proof of $\mathcal{F}$
satisfiable *Some $\mathcal{T}$-models of $\mathcal{A}$ (and there are some) satisfy some $\mathcal{C}_i$*	Model of $\mathcal{A}$ and some $\mathcal{C}_i$
contradictory-axioms *There are no $\mathcal{T}$-models of $\mathcal{A}$*	Refutation of $\mathcal{A}$
no-consequence *Some $\mathcal{T}$-models of $\mathcal{A}$ (and there are some) satisfy some $\mathcal{C}_i$, some satisfy $\overline{\mathcal{C}}$*	$\mathcal{T}$-model of $\mathcal{A}$ and some $\mathcal{C}_i$, $\mathcal{T}$-model of $\mathcal{A} \cup \overline{\mathcal{C}}$.
counter-satisfiable *Some $\mathcal{T}$-models of $\mathcal{A}$ (and there are some) satisfy $\overline{\mathcal{C}}$*	Model of $\mathcal{A} \cup \overline{\mathcal{C}}$
counter-theorem *All $\mathcal{T}$-models of $\mathcal{A}$ (and there are some) satisfy $\overline{\mathcal{C}}$*	Proof of $\overline{\mathcal{C}}$ from $\mathcal{A}$
counter-equivalent *$\mathcal{A}$ and $\overline{\mathcal{C}}$ have the same $\mathcal{T}$-models (and there are some)*	Proof of $\overline{\mathcal{C}}$ from $\mathcal{A}$ and proof of $\mathcal{A}$ from $\overline{\mathcal{C}}$
unsatisfiable-conclusion *All $\mathcal{T}$-interpretations satisfy $\overline{\mathcal{C}}$*	Proof of $\overline{\mathcal{C}}$
unsatisfiable *All $\mathcal{T}$-interpretations satisfy $\mathcal{A}$ and $\overline{\mathcal{C}}$*	Proof of $\neg\mathcal{F}$

Where $\mathcal{F}$ is an assertion whose FMP has assumption elements $\mathcal{A}_1, \ldots, \mathcal{A}_n$ and conclusion elements $\mathcal{C}_1, \ldots, \mathcal{C}_m$. Furthermore, let $\mathcal{A} := \{\mathcal{A}_1, \ldots, \mathcal{A}_n\}$ and $\mathcal{C} := \{\mathcal{C}_1, \ldots, \mathcal{C}_m\}$, and $\mathcal{F}^{-1}$ be the sequent that has the $\mathcal{C}_i$ as assumptions and the $\mathcal{A}_i$ as conclusions. Finally, let $\overline{\mathcal{C}} := \{\overline{\mathcal{C}_1}, \ldots, \overline{\mathcal{C}_m}\}$, where $\overline{\mathcal{C}_i}$ is a negation of $\mathcal{C}_i$.

Fig. 15.12. Proof status for assertions in a theory $\mathcal{T}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_F00188	150	■ 1.000	`\mathcal{A}:=\left\{\mathcal{A}_{1}, \ldots, \mathcal{A}_{n}\right\}`	$\mathcal{A}:=\left\{\mathcal{A}_{1}, \ldots, \mathcal{A}_{n}\right\}$	
kohlhase-omdoc_F00189	150	■ 1.000	`\mathcal{C}:=\left\{\mathcal{C}_{1}, \ldots, \mathcal{C}_{m}\right\}`	$\mathcal{C}:=\left\{\mathcal{C}_{1}, \ldots, \mathcal{C}_{m}\right\}$	

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status	just-by points to
tautology	Proof of $\mathcal{F}$
All $\mathcal{T}$-interpretations satisfy $\mathcal{A}$ and some $\mathcal{C}_i$	
tautologous-conclusion	Proof of $\mathcal{F}_c$.
All $\mathcal{T}$-interpretations satisfy some $\mathcal{C}_j$	
equivalent	Proofs of $\mathcal{F}$ and $\mathcal{F}^{-1}$
$\mathcal{A}$ and $\mathcal{C}$ have the same $\mathcal{T}$-models (and there are some)	
theorem	Proof of $\mathcal{F}$
All $\mathcal{T}$-models of $\mathcal{A}$ (and there are some) satisfy some $\mathcal{C}_i$	
satisfiable	Model of $\mathcal{A}$ and some $\mathcal{C}_i$
Some $\mathcal{T}$-models of $\mathcal{A}$ (and there are some) satisfy some $\mathcal{C}_i$	
contradictory-axioms	Refutation of $\mathcal{A}$
There are no $\mathcal{T}$-models of $\mathcal{A}$	
no-consequence	$\mathcal{T}$-model of $\mathcal{A}$ and some $\mathcal{C}_i$, $\mathcal{T}$-model of $\mathcal{A}\cup\overline{\mathcal{C}}$.
Some $\mathcal{T}$-models of $\mathcal{A}$ (and there are some) satisfy some $\mathcal{C}_i$, some satisfy $\overline{\mathcal{C}}$	
counter-satisfiable	Model of $\mathcal{A}\cup\overline{\mathcal{C}}$
Some $\mathcal{T}$-models of $\mathcal{A}$ (and there are some) satisfy $\overline{\mathcal{C}}$	
counter-theorem	Proof of $\overline{\mathcal{C}}$ from $\mathcal{A}$
All $\mathcal{T}$-models of $\mathcal{A}$ (and there are some) satisfy $\overline{\mathcal{C}}$	
counter-equivalent	Proof of $\overline{\mathcal{C}}$ from $\mathcal{A}$ and proof of $\mathcal{A}$ from $\overline{\mathcal{C}}$
$\mathcal{A}$ and $\overline{\mathcal{C}}$ have the same $\mathcal{T}$-models (and there are some)	
unsatisfiable-conclusion	Proof of $\overline{\mathcal{C}}$
All $\mathcal{T}$-interpretations satisfy $\overline{\mathcal{C}}$	
unsatisfiable	Proof of $\neg\mathcal{F}$
All $\mathcal{T}$-interpretations satisfy $\mathcal{A}$ and $\overline{\mathcal{C}}$	

Where $\mathcal{F}$ is an assertion whose FMP has assumption elements $\mathcal{A}_1,\ldots,\mathcal{A}_n$ and conclusion elements $\mathcal{C}_1,\ldots,\mathcal{C}_m$. Furthermore, let $\mathcal{A}:=\{\mathcal{A}_1,\ldots,\mathcal{A}_n\}$ and $\mathcal{C}:=\{\mathcal{C}_1,\ldots,\mathcal{C}_m\}$, and $\mathcal{F}^{-1}$ be the sequent that has the $\mathcal{C}_i$ as assumptions and the $\mathcal{A}_i$ as conclusions. Finally, let $\overline{\mathcal{C}}:=\{\overline{\mathcal{C}_1},\ldots,\overline{\mathcal{C}_m}\}$, where $\overline{\mathcal{C}_i}$ is a negation of $\mathcal{C}_i$.

Fig. 15.12. Proof status for assertions in a theory $\mathcal{T}$

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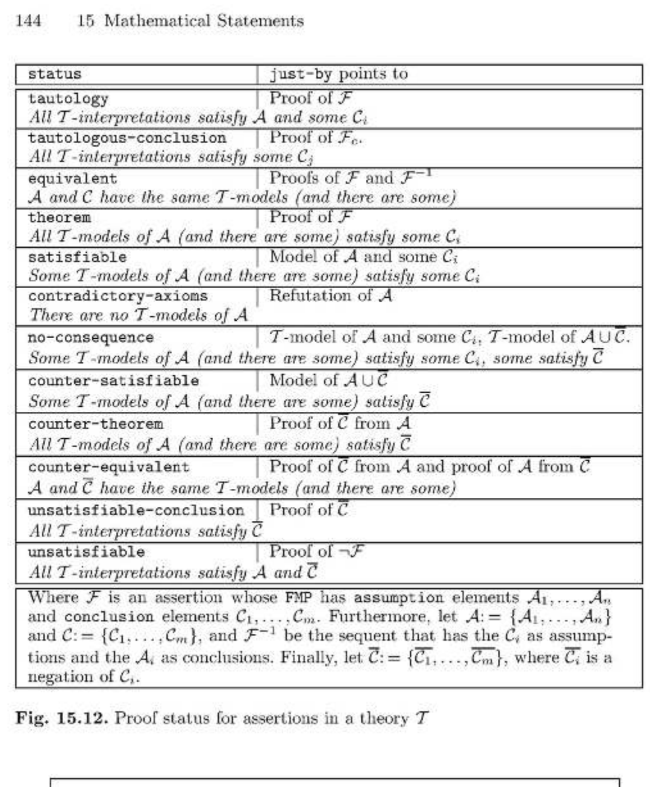
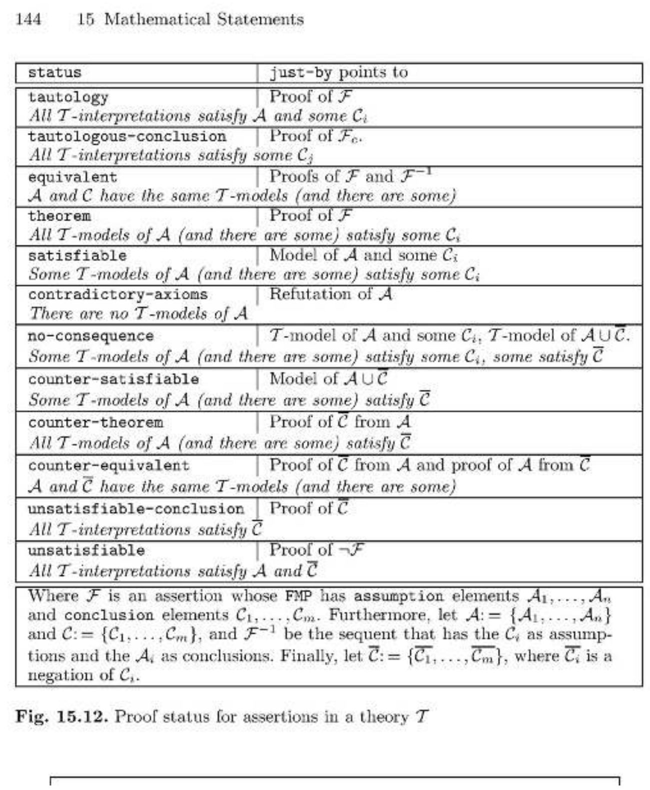
(Image 2 shows the same excerpt as image 1: the table of status / just-by points to, the explanatory paragraph and the caption Fig. 15.12. Proof status for assertions in a theory $\mathcal{T}$)

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_F00190	150	■ 1.000	`\mathcal{A}_{i}`	$\mathcal{A}_{i}$	
kohlhase-omdoc_F00191	150	■ 1.000	`\overline{\mathcal{C}}:=\left\{\overline{\mathcal{C}_{1}}, \ldots, \overline{\mathcal{C}_{m}}\right\}`	$\overline{\mathcal{C}}:=\left\{\overline{\mathcal{C}_{1}}, \ldots, \overline{\mathcal{C}_{m}}\right\}$	

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status	just-by points to
tautology	Proof of $\mathcal{F}$
All $\mathcal{T}$-interpretations satisfy $\mathcal{A}$ and some $\mathcal{C}_i$	
tautologous-conclusion	Proof of $\mathcal{F}_c$.
All $\mathcal{T}$-interpretations satisfy some $\mathcal{C}_j$	
equivalent	Proofs of $\mathcal{F}$ and $\mathcal{F}^{-1}$
$\mathcal{A}$ and $\mathcal{C}$ have the same $\mathcal{T}$-models (and there are some)	
theorem	Proof of $\mathcal{F}$
All $\mathcal{T}$-models of $\mathcal{A}$ (and there are some) satisfy some $\mathcal{C}_i$	
satisfiable	Model of $\mathcal{A}$ and some $\mathcal{C}_i$
Some $\mathcal{T}$-models of $\mathcal{A}$ (and there are some) satisfy some $\mathcal{C}_i$	
contradictory-axioms	Refutation of $\mathcal{A}$
There are no $\mathcal{T}$-models of $\mathcal{A}$	
no-consequence	$\mathcal{T}$-model of $\mathcal{A}$ and some $\mathcal{C}_i$, $\mathcal{T}$-model of $\mathcal{A}\cup\overline{\mathcal{C}}$.
Some $\mathcal{T}$-models of $\mathcal{A}$ (and there are some) satisfy some $\mathcal{C}_i$, some satisfy $\overline{\mathcal{C}}$	
counter-satisfiable	Model of $\mathcal{A}\cup\overline{\mathcal{C}}$
Some $\mathcal{T}$-models of $\mathcal{A}$ (and there are some) satisfy $\overline{\mathcal{C}}$	
counter-theorem	Proof of $\overline{\mathcal{C}}$ from $\mathcal{A}$
All $\mathcal{T}$-models of $\mathcal{A}$ (and there are some) satisfy $\overline{\mathcal{C}}$	
counter-equivalent	Proof of $\overline{\mathcal{C}}$ from $\mathcal{A}$ and proof of $\mathcal{A}$ from $\overline{\mathcal{C}}$
$\mathcal{A}$ and $\overline{\mathcal{C}}$ have the same $\mathcal{T}$-models (and there are some)	
unsatisfiable-conclusion	Proof of $\overline{\mathcal{C}}$
All $\mathcal{T}$-interpretations satisfy $\overline{\mathcal{C}}$	
unsatisfiable	Proof of $\neg\mathcal{F}$
All $\mathcal{T}$-interpretations satisfy $\mathcal{A}$ and $\overline{\mathcal{C}}$	

Where $\mathcal{F}$ is an assertion whose FMP has assumption elements $\mathcal{A}_1, \ldots, \mathcal{A}_n$ and conclusion elements $\mathcal{C}_1, \ldots, \mathcal{C}_m$. Furthermore, let $\mathcal{A} := \{\mathcal{A}_1, \ldots, \mathcal{A}_n\}$ and $\mathcal{C} := \{\mathcal{C}_1, \ldots, \mathcal{C}_m\}$, and $\mathcal{F}^{-1}$ be the sequent that has the $\mathcal{C}_i$ as assumptions and the $\mathcal{A}_i$ as conclusions. Finally, let $\overline{\mathcal{C}} := \{\overline{\mathcal{C}_1}, \ldots, \overline{\mathcal{C}_m}\}$, where $\overline{\mathcal{C}_i}$ is a negation of $\mathcal{C}_i$.

Fig. 15.12. Proof status for assertions in a theory $\mathcal{T}$

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(Image 2 reproduces the same page: status / just-by points to table and Fig. 15.12. Proof status for assertions in a theory $\mathcal{T}$)

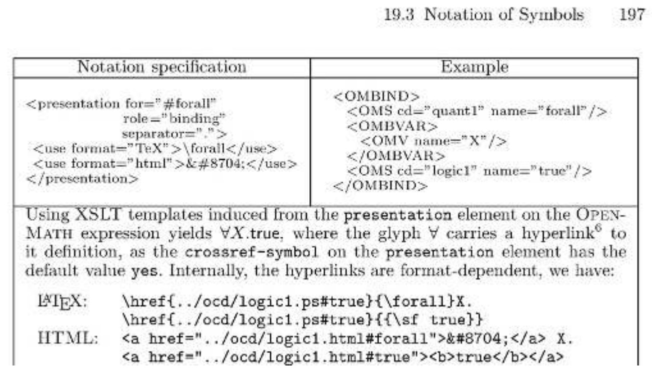
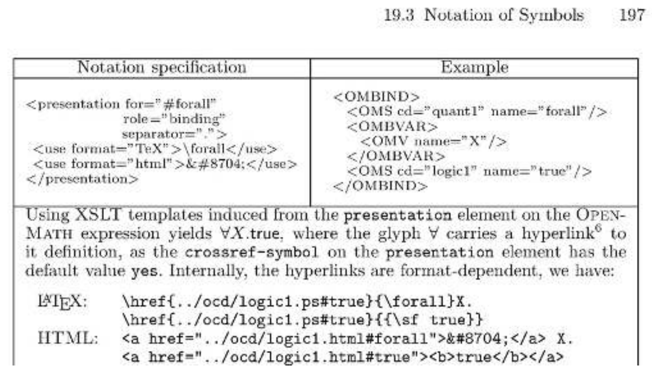
Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_F00192	150	1.000	`\overline{\mathcal{C}_{i}}`	$\overline{\mathcal{C}_{i}}$	144 15 Mathematical Statements status | just-by points to tautology | Proof of $\mathcal{F}$ *All $\mathcal{T}$-interpretations satisfy $\mathcal{A}$ and some $\mathcal{C}_i$* tautologous-conclusion | Proof of $\mathcal{F}_c$. *All $\mathcal{T}$-interpretations satisfy some $\mathcal{C}_j$* equivalent | Proofs of $\mathcal{F}$ and $\mathcal{F}^{-1}$ *$\mathcal{A}$ and $\mathcal{C}$ have the same $\mathcal{T}$-models (and there are some)* theorem | Proof of $\mathcal{F}$ *All $\mathcal{T}$-models of $\mathcal{A}$ (and there are some) satisfy some $\mathcal{C}_i$* satisfiable | Model of $\mathcal{A}$ and some $\mathcal{C}_i$ *Some $\mathcal{T}$-models of $\mathcal{A}$ (and there are some) satisfy some $\mathcal{C}_i$* contradictory-axioms | Refutation of $\mathcal{A}$ *There are no $\mathcal{T}$-models of $\mathcal{A}$* no-consequence | $\mathcal{T}$-model of $\mathcal{A}$ and some $\mathcal{C}_i$, $\mathcal{T}$-model of $\mathcal{A}\cup\overline{\mathcal{C}}$. *Some $\mathcal{T}$-models of $\mathcal{A}$ (and there are some) satisfy some $\mathcal{C}_i$, some satisfy $\overline{\mathcal{C}}$* counter-satisfiable | Model of $\mathcal{A}\cup\overline{\mathcal{C}}$ *Some $\mathcal{T}$-models of $\mathcal{A}$ (and there are some) satisfy $\overline{\mathcal{C}}$* counter-theorem | Proof of $\overline{\mathcal{C}}$ from $\mathcal{A}$ *All $\mathcal{T}$-models of $\mathcal{A}$ (and there are some) satisfy $\overline{\mathcal{C}}$* counter-equivalent | Proof of $\overline{\mathcal{C}}$ from $\mathcal{A}$ and proof of $\mathcal{A}$ from $\overline{\mathcal{C}}$ *$\mathcal{A}$ and $\overline{\mathcal{C}}$ have the same $\mathcal{T}$-models (and there are some)* unsatisfiable-conclusion | Proof of $\overline{\mathcal{C}}$ *All $\mathcal{T}$-interpretations satisfy $\overline{\mathcal{C}}$* unsatisfiable | Proof of $\neg\mathcal{F}$ *All $\mathcal{T}$-interpretations satisfy $\mathcal{A}$ and $\overline{\mathcal{C}}$* Where $\mathcal{F}$ is an assertion whose FMP has assumption elements $\mathcal{A}_1,\ldots,\mathcal{A}_n$ and conclusion elements $\mathcal{C}_1,\ldots,\mathcal{C}_m$. Furthermore, let $\mathcal{A}:=\{\mathcal{A}_1,\ldots,\mathcal{A}_n\}$ and $\mathcal{C}:=\{\mathcal{C}_1,\ldots,\mathcal{C}_m\}$, and $\mathcal{F}^{-1}$ be the sequent that has the $\mathcal{C}_i$ as assumptions and the $\mathcal{A}_i$ as conclusions. Finally, let $\overline{\mathcal{C}}:=\{\overline{\mathcal{C}_1},\ldots,\overline{\mathcal{C}_m}\}$, where $\overline{\mathcal{C}_i}$ is a negation of $\mathcal{C}_i$. **Fig. 15.12.** Proof status for assertions in a theory $\mathcal{T}$
kohlhase-omdoc_F00193	152	0.997	`\mathcal{E}`	$\mathcal{E}$	If $\mathcal{E}$ is an example for a conjecture **C**, then we have to consider the situa-
kohlhase-omdoc_F00194	152	0.997	`(\mathcal{W}, \mathbf{A})`	$(\mathcal{W}, \mathbf{A})$	If $\mathcal{E}$ is an example for a conjecture **C**, then we have to consider the situa-
kohlhase-omdoc_F00195	152	0.992	`\mathcal{W}=\left(\mathcal{W}_{1}, \ldots, \mathcal{W}_{n}\right)`	$\mathcal{W}=\left(\mathcal{W}_{1}, \ldots, \mathcal{W}_{n}\right)$	where $\mathcal{Q}$ is a sequence $\mathcal{Q}_1W_1,\ldots,\mathcal{Q}_mW_m$ of $m \geq n = \#\mathcal{W}$ quantifications of
kohlhase-omdoc_F00196	152	0.923	`\mathbf{S}`	$\mathbf{S}$	tifiers of $\mathcal{Q}$ and $\mathbf{D}'$ be $\mathbf{D}$ only that all the W_{i_j} such that the $\mathcal{Q}_{i_j}$ are
kohlhase-omdoc_F00197	152	0.923	`\mathbf{S}\left(\mathcal{W}_{1}, \ldots, \mathcal{W}_{n}\right)`	$\mathbf{S}\left(\mathcal{W}_{1}, \ldots, \mathcal{W}_{n}\right)$	tifiers of $\mathcal{Q}$ and $\mathbf{D}'$ be $\mathbf{D}$ only that all the W_{i_j} such that the $\mathcal{Q}_{i_j}$ are
kohlhase-omdoc_F00198	152	0.999	`\mathbf{C}`	$\mathbf{C}$	OMDoc specifies this intuition in an example element that contains a
kohlhase-omdoc_F00199	152	0.999	`\mathcal{Q} \mathbf{D}`	$\mathcal{Q} \mathbf{D}$	OMDoc specifies this intuition in an example element that contains a
kohlhase-omdoc_F00200	152	0.999	`\mathbf{D}`	$\mathbf{D}$	OMDoc specifies this intuition in an example element that contains a
kohlhase-omdoc_F00201	152	1.000	`\mathcal{Q}`	$\mathcal{Q}$	witnesses). It has the attributes
kohlhase-omdoc_F00202	152	1.000	`\mathcal{Q}_{1} W_{1}, \ldots, \mathcal{Q}_{m} W_{m}`	$\mathcal{Q}_{1} W_{1}, \ldots, \mathcal{Q}_{m} W_{m}$	witnesses). It has the attributes
kohlhase-omdoc_F00203	152	1.000	`m \geq n=\# \mathcal{W}`	$m \geq n=\# \mathcal{W}$	witnesses). It has the attributes
kohlhase-omdoc_F00204	152	0.999	`\mathcal{Q}_{i}`	$\mathcal{Q}_{i}$	for specifying for which concepts or assertions it is an example. This is
kohlhase-omdoc_F00205	152	0.999	`\forall`	$\forall$	for specifying for which concepts or assertions it is an example. This is
kohlhase-omdoc_F00206	152	0.999	`\exists`	$\exists$	for specifying for which concepts or assertions it is an example. This is

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_F00207	152	0.999	`\mathcal{Q}^{\prime}`	$\mathcal{Q}^{\prime}$	for specifying for which concepts or assertions it is an example. This is
kohlhase-omdoc_F00208	152	0.999	`m-n`	$m-n$	for specifying for which concepts or assertions it is an example. This is
kohlhase-omdoc_F00209	152	0.985	`\mathbf{D}^{\prime}`	$\mathbf{D}^{\prime}$	a reference to a whitespace-separated list of URI references to symbol,
kohlhase-omdoc_F00210	152	0.985	`W_{i_{j}}`	$W_{i_{j}}$	a reference to a whitespace-separated list of URI references to symbol,
kohlhase-omdoc_F00211	152	0.985	`\mathcal{Q}_{i_{j}}`	$\mathcal{Q}_{i_{j}}$	a reference to a whitespace-separated list of URI references to symbol,
kohlhase-omdoc_F00212	152	0.998	`\mathcal{W}_{j}`	$\mathcal{W}_{j}$	definition, or assertion elements.
kohlhase-omdoc_F00213	152	0.998	`1 \leq j \leq n`	$1 \leq j \leq n$	definition, or assertion elements.
kohlhase-omdoc_F00214	152	0.998	`\mathcal{E}=(\mathcal{W}, \mathbf{A})`	$\mathcal{E}=(\mathcal{W}, \mathbf{A})$	definition, or assertion elements.
kohlhase-omdoc_F00215	152	1.000	`\mathbf{A}=\mathcal{Q}^{\prime} \mathbf{D}^{\prime}`	$\mathbf{A}=\mathcal{Q}^{\prime} \mathbf{D}^{\prime}$	type specifying the aspect, the value is one of for or against
kohlhase-omdoc_F00216	152	1.000	`\mathbf{A}=\neg \mathcal{Q}^{\prime} \mathbf{D}^{\prime}`	$\mathbf{A}=\neg \mathcal{Q}^{\prime} \mathbf{D}^{\prime}$	type specifying the aspect, the value is one of for or against
kohlhase-omdoc_F00217	154	1.000	`A^{*}`	A^{*}	148 15 Mathematical Statements
kohlhase-omdoc_F00218	154	0.914	`\epsilon`	ϵ	In Listing 15.15 we show an example of the usage of an example element
kohlhase-omdoc_F00219	154	0.914	`\mathcal{W}=\left(A^{*},::, \epsilon\right)`	$\mathcal{W}=\left(A^{*},::, \epsilon\right)$	In Listing 15.15 we show an example of the usage of an example element
kohlhase-omdoc_F00220	154	1.000	`\left(\mathcal{W}, \mathbf{A}^{\prime}\right)`	$\left(\mathcal{W}, \mathbf{A}^{\prime}\right)$	with xml:id="mon.ex1" in Listing 15.15 is an example for the concept of a monoid, since it encodes the pair $(\mathcal{W}, \mathbf{A})$ where $\mathbf{A}$ is given by reference to
kohlhase-omdoc_F00221	154	0.380	`M,`	$M,$	after the inline definition. The real problem is that the inline definition is linguistically a phrase-level construct, while the omtext element is a discourse-
kohlhase-omdoc_F00222	156	0.939	`(\langle\langle t o p+t h c\rangle\rangle) *`	$(\langle\langle t o p+t h c\rangle\rangle) *$	150 15 Mathematical Statements Element | Attributes | D | Content
kohlhase-omdoc_F00223	161	1.000	`\emptyset`	$\emptyset$	
kohlhase-omdoc_F00224	161	1.000	`\iota`	ι	
kohlhase-omdoc_F00225	161	1.000	`\{a\}`	$\{a\}$	
kohlhase-omdoc_F00226	161	1.000	`\iota(a, \emptyset)=\iota(a, \iota(a, \emptyset))`	$\iota(a, \emptyset)=\iota(a, \iota(a, \emptyset))$	
kohlhase-omdoc_F00227	162	1.000	`\mathbb{P}`	$\mathbb{P}$	A sortdef element is a highly condensed piece of syntax that declares a
kohlhase-omdoc_F00228	166	1.000	`P`	P	oof steps for referencing convenience. Figure 17.1 will be used as a running
kohlhase-omdoc_F00229	166	1.000	`P=\left\{p_{1}, \ldots, p_{n}\right\}`	$P=\left\{p_{1}, \ldots, p_{n}\right\}$	1. We proceed by assuming that P is finite and rea
kohlhase-omdoc_F00230	166	1.000	`p_{i}`	p_{i}	1. We proceed by assuming that P is finite and rea
kohlhase-omdoc_F00231	166	1.000	`q \stackrel{\text { def }}{=} p_{1} \cdots p_{n}+1`	$q \stackrel{\text { def }}{=} p_{1} \cdots p_{n}+1$	contradiction. 2. Let P be finite.
kohlhase-omdoc_F00232	166	1.000	`p_{i} \in P`	$p_{i} \in P$	3. Then $P=\{p_1,\ldots,p_n\}$ for some p_i.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_F00233	166	1.000	`q>p_{i}`	$q>p_{i}$	3. Then $P = \{p_1, \ldots, p_n\}$ for some p_i.
kohlhase-omdoc_F00234	166	1.000	`q \notin P`	$q \notin P$	3. Then $P = \{p_1, \ldots, p_n\}$ for some p_i.
kohlhase-omdoc_F00235	166	1.000	`q`	q	4. Let $q \stackrel{\text{def}}{=} p_1 \cdots p_n + 1$.
kohlhase-omdoc_F00236	166	1.000	`q=p_{i} k+1`	$q=p_{i} k+1$	6. We prove the absurdity by showing that q is prime:
kohlhase-omdoc_F00237	166	1.000	`k`	k	For each $p_i \in P$ we have $q = p_i k + 1$ for sor
kohlhase-omdoc_F00238	166	0.990	`\bar{\lambda} \mu \tilde{\mu}`	$\bar{\lambda} \mu \tilde{\mu}$	up this aspect. Even if the proof in Figure 17.1 is very short and simple, we
kohlhase-omdoc_F00239	168	1.000	`12 f f`	$12 f f$	and an optional type to single out special cases of proofs steps.
kohlhase-omdoc_F00240	170	0.986	`\left[\mathrm{BCF}^{+} 97\right]`	$\left[\mathrm{BCF}^{+} 97\right]$	This object is an alternative representation of certain proofs, see Section 17.4.
kohlhase-omdoc_F00241	179	0.791	`(R,+)`	$(R,+)$	elementary algebraic hierarchy: for a theory of rings, we should be able to
kohlhase-omdoc_F00242	179	1.000	`\left(R^{*}, *\right)`	$\left(R^{*}, *\right)$	inherit the additive group structure from the theory group of groups and the
kohlhase-omdoc_F00243	179	1.000	`R^{*}:=\{r \in R \mid r \neq 0\}`	$R^{*}:=\{r \in R \mid r \neq 0\}$	structure of a multiplicative monoid from the theory monoid: A ring is a set R together with two operations $+$ and $*$, such that $(R,+)$ is a group with
kohlhase-omdoc_F00244	179	0.604	`M=(S, \circ, e)`	$M=(S, \circ, e)$	
kohlhase-omdoc_F00245	179	1.000	`S \subseteq M`	$S \subseteq M$	
kohlhase-omdoc_F00246	179	1.000	`S^{*}:=\{r \in S r \neq e\}`	$S^{*}:=\{r \in S r \neq e\}$	
kohlhase-omdoc_F00247	180	1.000	`{ }^{3} t_{1}, \ldots, t_{n}`	${ }^{3} t_{1}, \ldots, t_{n}$	
kohlhase-omdoc_F00248	180	1.000	`\sigma: \mathcal{S} \rightarrow \mathcal{T}`	$\sigma: \mathcal{S} \rightarrow \mathcal{T}$	
kohlhase-omdoc_F00249	180	1.000	`\sigma\left(t_{i}\right)`	$\sigma\left(t_{i}\right)$	constitutive statements $\sigma(t_i)$ from $\mathcal{S}$. For instance, the theory of rings inherits
kohlhase-omdoc_F00250	180	1.000	`\forall x . x+0=x`	$\forall x . x+0=x$	the axiom $\forall x.x + 0 = x$ from the theory of monoids as $\sigma(\forall x.x + 0 = x) =$
kohlhase-omdoc_F00251	180	1.000	`\sigma(\forall x . x+0=x)=`	$\sigma(\forall x . x+0=x)=$	the axiom $\forall x.x + 0 = x$ from the theory of monoids as $\sigma(\forall x.x + 0 = x) =$
kohlhase-omdoc_F00252	180	0.999	`\forall x . x * 1=x`	$\forall x . x * 1=x$	To speci
kohlhase-omdoc_F00253	181	0.999	`R,+, 0,-, *, 1`	$R,+, 0,-, *, 1$	the morphism. In our example, the symbols R, $+$, 0, $-$, $*$, 1 are visible in the
kohlhase-omdoc_F00254	181	1.000	`L`	L	order to refrain from polluting the name space.
kohlhase-omdoc_F00255	181	1.000	`L^{\prime}`	$L^{\prime}$	As morphisms often contain common prefixes, the morphism element has
kohlhase-omdoc_F00256	182	0.879	`\sigma_{i}`	σ_{i}	18.2 Postulated Theory Inclusions 177
kohlhase-omdoc_F00257	182	0.879	`\sigma_{i-1}`	σ_{i-1}	18.2 Postulated Theory Inclusions 177
kohlhase-omdoc_F00258	182	0.902	`\mathcal{S}, \sigma(\mathbf{S})`	$\mathcal{S}, \sigma(\mathbf{S})$	In OMDoc, we weaken this logical property to a structural one: We say
kohlhase-omdoc_F00259	182	0.902	`\sigma(\mathbf{S})`	$\sigma(\mathbf{S})$	In OMDoc, we weaken this logical property to a structural one: We say

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_F00260	182	0.905	`\mathbf{T}`	$\mathbf{T}$	lated mathematical vernacular. In this view, a **structural theory inclusion**
kohlhase-omdoc_F00261	182	1.000	`\mathcal{T}^{5}`	$\mathcal{T}^{5}$	[4] Mathematical objects that can be represented using the only symbols of the
kohlhase-omdoc_F00262	183	1.000	`\sigma(\mathbf{A})`	$\sigma(\mathbf{A})$	We call a theory inclusion $\sigma{:}\mathcal{S} \to \mathcal{T}$ **conservative**, iff **A** is already a
kohlhase-omdoc_F00263	183	0.421	`>\forall x, y, z \in M . z \circ(y \circ x)=(z \circ y) \circ x</`	$> \forall x, y, z \in M . z \circ(y \circ x)=(z \circ y) \circ x</$	
kohlhase-omdoc_F00264	183	0.421	`>`	$>$	
kohlhase-omdoc_F00265	183	0.765	`\forall x, y \in M . y \circ x \in M</`	$\forall x, y \in M . y \circ x \in M</$	
kohlhase-omdoc_F00266	183	0.325	`\forall x \in M . e \circ x=x<`	$\forall x \in M . e \circ x=x<$	
kohlhase-omdoc_F00267	186	0.998	`\left[\mathrm{BCD}^{+} 02\right]`	$\left[\mathrm{BCD}^{+} 02\right]$	morphism can alleviate both.
kohlhase-omdoc_F00268	187	1.000	`\Gamma`	Γ	assume for the sake of this argument that for $\mathcal{X}_i \in \{\mathcal{A},\mathcal{B},\mathcal{C}\}$ theories $\mathcal{X}_1$ and
kohlhase-omdoc_F00269	187	1.000	`\mathcal{T}_{1}`	$\mathcal{T}_{1}$	assume for the sake of this argument that for $\mathcal{X}_i \in \{\mathcal{A},\mathcal{B},\mathcal{C}\}$ theories $\mathcal{X}_1$ and
kohlhase-omdoc_F00270	187	1.000	`\mathcal{T}_{2}`	$\mathcal{T}_{2}$	assume for the sake of this argument that for $\mathcal{X}_i \in \{\mathcal{A},\mathcal{B},\mathcal{C}\}$ theories $\mathcal{X}_1$ and
kohlhase-omdoc_F00271	187	1.000	`\mathcal{C}_{1}`	$\mathcal{C}_{1}$	assume for the sake of this argument that for $\mathcal{X}_i \in \{\mathcal{A},\mathcal{B},\mathcal{C}\}$ theories $\mathcal{X}_1$ and
kohlhase-omdoc_F00272	187	0.999	`\mathcal{A}_{1}`	$\mathcal{A}_{1}$	$\mathcal{X}_2$ are so similar that axiom inclusions (they are indicated by thin dashed
kohlhase-omdoc_F00273	187	0.999	`\mathcal{B}_{1}`	$\mathcal{B}_{1}$	$\mathcal{X}_2$ are so similar that axiom inclusions (they are indicated by thin dashed
kohlhase-omdoc_F00274	187	0.999	`\mathcal{C}_{2}`	$\mathcal{C}_{2}$	$\mathcal{X}_2$ are so similar that axiom inclusions (they are indicated by thin dashed
kohlhase-omdoc_F00275	187	0.999	`\mathcal{A}_{2}`	$\mathcal{A}_{2}$	$\mathcal{X}_2$ are so similar that axiom inclusions (they are indicated by thin dashed
kohlhase-omdoc_F00276	187	1.000	`\mathcal{B}_{2}`	$\mathcal{B}_{2}$	arrows in Figure 18.8 and have the formula-mappings α, β, and γ) are easy to prove[8]
kohlhase-omdoc_F00277	187	1.000	`\mathcal{X}_{i} \in\{\mathcal{A}, \mathcal{B}, \mathcal{C}\}`	$\mathcal{X}_{i} \in\{\mathcal{A}, \mathcal{B}, \mathcal{C}\}$	To justify Γ, we must prove that the Γ-translations of all the theory-
kohlhase-omdoc_F00278	187	1.000	`\mathcal{X}_{1}`	$\mathcal{X}_{1}$	To justify Γ, we must prove that the Γ-translations of all the theory-
kohlhase-omdoc_F00279	187	1.000	`\mathcal{X}_{2}`	$\mathcal{X}_{2}$	constitutive statements of $\mathcal{C}_1$ are provable in $\mathcal{C}_2$. So let statement **B** be theory-
kohlhase-omdoc_F00280	187	0.995	`\alpha, \beta`	α, β	provable in $\mathcal{B}_2$ since β is an axiom inclusion. Moreover, we know that $\sigma(\beta(\mathbf{B}))$
kohlhase-omdoc_F00281	187	0.995	`\gamma`	γ	provable in $\mathcal{B}_2$ since β is an axiom inclusion. Moreover, we know that $\sigma(\beta(\mathbf{B}))$
kohlhase-omdoc_F00282	187	1.000	`\beta(\mathbf{B})`	$\beta(\mathbf{B})$	inclusions and checking translation compatibility.
kohlhase-omdoc_F00283	187	1.000	`\beta`	β	We will call a situation, where a theory $\mathcal{T}$ can be reached by an axiom inclusion with a subsequent chain of theory inclusions a **local chain** (with
kohlhase-omdoc_F00284	187	1.000	`\sigma(\beta(\mathbf{B}))`	$\sigma(\beta(\mathbf{B}))$	We will call a situation, where a theory $\mathcal{T}$ can be reached by an axiom inclusion with a subsequent chain of theory inclusions a **local chain** (with
kohlhase-omdoc_F00285	187	1.000	`\Gamma=\sigma \circ \beta`	$\Gamma=\sigma \circ \beta$	[8] A common source of situations like this is where the $\mathcal{X}_2$ are variants of the $\mathcal{X}_1$
kohlhase-omdoc_F00286	187	1.000	`\Gamma^{-1}`	Γ^{-1}	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_F00287	188	1.000	\tau:=\sigma_{n} \circ \cdots \circ \sigma_{1} \circ \sigma	$\tau:=\sigma_{n} \circ \cdots \circ \sigma_{1} \circ \sigma$	
kohlhase-omdoc_F00288	188	1.000	\mathcal{S} \xrightarrow{\sigma} \mathcal{T}_{1}	$\mathcal{S} \xrightarrow{\sigma} \mathcal{T}_{1}$	
kohlhase-omdoc_F00289	188	1.000	\mathcal{T}_{i} \xrightarrow{\sigma_{i}} \mathcal{T}_{i+1}	$\mathcal{T}_{i} \xrightarrow{\sigma_{i}} \mathcal{T}_{i+1}$	morphism $\tau:=\sigma_n \circ \cdots \circ \sigma_1 \circ \sigma$), if $\mathcal{S} \xrightarrow{\sigma} \mathcal{T}_1$ is an axiom inclusion or (local
kohlhase-omdoc_F00290	188	1.000	\mathcal{T}_{1}, \mathcal{T}_{2}, \mathcal{A}_{1}	$\mathcal{T}_{1}, \mathcal{T}_{2}, \mathcal{A}_{1}$	$\mathcal{T}_2 \longrightarrow \mathcal{B}_2 \xrightarrow{q} \mathcal{C}_2$ $\mathcal{B}_1 \overset{\beta}{\dashrightarrow} \mathcal{B}_2 \xrightarrow{q} \mathcal{C}_2$
kohlhase-omdoc_F00291	188	1.000	\mathcal{X}	$\mathcal{X}$	We call a situation where we have a formula mapping $\mathcal{S} \longrightarrow \mathcal{T}$, and an axiom inclusion $\mathcal{X} \xrightarrow{\sigma_{\mathcal{X}}} \mathcal{T}$ for every theory $\mathcal{X}$ that $\mathcal{S}$ inherits from a
kohlhase-omdoc_F00292	188	1.000	\mathcal{S} \xrightarrow{\sigma} \mathcal{T}	$\mathcal{S} \xrightarrow{\sigma} \mathcal{T}$	indeed a theory inclusion.
kohlhase-omdoc_F00293	188	1.000	\mathcal{X} \xrightarrow{\sigma_{\mathcal{X}}} \mathcal{T}	$\mathcal{X} \xrightarrow{\sigma_{\mathcal{X}}} \mathcal{T}$	[9] Note for the leftmost two chains use the fact that theory inclusions (in our case definitional ones) are also axiom inclusions by definition.
kohlhase-omdoc_F00294	188	1.000	\sigma_{\mathcal{X}}	$\sigma_{\mathcal{X}}$	
kohlhase-omdoc_F00295	190	0.996	i	i	inclusion) and the attribute globals attribute, which contains a white-
kohlhase-omdoc_F00296	190	1.000	\sigma_{1}	σ_{1}	space-separated list of URI references to theory
kohlhase-omdoc_F00297	190	1.000	\sigma_{2}	σ_{2}	space-separated list of URI references to theory
kohlhase-omdoc_F00298	197	1.000	\forall P, Q . P \vee Q \Rightarrow Q \vee P	$\forall P, Q . P \vee Q \Rightarrow Q \vee P$	5 <OMS cd="logic1" name="implies"/> <OMA><OMS cd="logic1" name="or"/>
kohlhase-omdoc_F00299	199	1.000	g c d	gcd	gcd in English but ggT ("größter gemeinsamer Teiler") in German). Obvi-
kohlhase-omdoc_F00300	199	1.000	g g T	ggT	gcd in English but ggT ("größter gemeinsamer Teiler") in German). Obvi-
kohlhase-omdoc_F00301	200	1.000	f^{\prime}	$f^{\prime}$	the arguments, (this is the generic mathematical function notation). For postfix the function is put behind the arguments, e.g. for derivatives:
kohlhase-omdoc_F00302	200	1.000	(x+2)^{2}	$(x+2)^{2}$	<OMV name="x"/> <OMV name="y"/> <OMS cd="arith1" name="pow- <OMV name="y"/>
kohlhase-omdoc_F00303	200	1.000	x+y^{2}	$x+y^{2}$	<OMV name="x"/> <OMV name="y"/> <OMS cd="arith1" name="pow- <OMV name="y"/>
kohlhase-omdoc_F00304	201	0.952	*, \wedge, \cap	$*, \wedge, \cap$	ation and Presentation Precedence / Operators / Comment 200 / +,- / unary 200 / ^ / exponentiation
kohlhase-omdoc_F00305	201	0.952	=, \neq, \leq,<,>, \geq	$=, \neq, \leq,<,>, \geq$	ation and Presentation Precedence / Operators / Comment 200 / +,- / unary 200 / ^ / exponentiation
kohlhase-omdoc_F00306	201	1.000	(a, b)	(a, b)	the default value – means cross-referencing only on the print-form of the
kohlhase-omdoc_F00307	201	0.912	\mathrm{IAT}_{\mathrm{E}} \mathrm{X}	$\mathrm{IAT}_{\mathrm{E}} \mathrm{X}$	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_F00308 **[conf 0.000]**	202	■0.000	`\forall X`	$\forall X$	 19.3 Notation of Symbols 197 Notation specification / Example <presentation for="#forall" role="binding" separator="."> <use format="TeX">\forall</use> <use format="html">∀</use> </presentation> <OMBIND> <OMS cd="quant1" name="forall"/> <OMBVAR> <OMV name="X"/> </OMBVAR> <OMS cd="logic1" name="true"/> </OMBIND> Using XSLT templates induced from the **presentation** element on the OPEN-MATH expression yields ∀X.true, where the glyph ∀ carries a hyperlink[6] to it definition, as the **crossref-symbol** on the **presentation** element has the default value **yes**. Internally, the hyperlinks are format-dependent, we have: LaTeX: `\href{../ocd/logic1.ps#true}{\forall}X.` `\href{../ocd/logic1.ps#true}{{\sf true}}` HTML: `<a href="../ocd/logic1.html#forall">∀</a> X.` `<a href="../ocd/logic1.html#true"><b>true</b></a>`
kohlhase-omdoc_F00309 **[conf 0.000]**	202	■0.000	`{ }^{\varnothing}`	$\varnothing$	 19.3 Notation of Symbols 197 Notation specification / Example <presentation for="#forall" role="binding" separator="."> <use format="TeX">\forall</use> <use format="html">∀</use> </presentation> <OMBIND> <OMS cd="quant1" name="forall"/> <OMBVAR> <OMV name="X"/> </OMBVAR> <OMS cd="logic1" name="true"/> </OMBIND> Using XSLT templates induced from the **presentation** element on the OPEN-MATH expression yields ∀X.true, where the glyph ∀ carries a hyperlink[6] to it definition, as the **crossref-symbol** on the **presentation** element has the default value **yes**. Internally, the hyperlinks are format-dependent, we have: LaTeX: `\href{../ocd/logic1.ps#true}{\forall}X.` `\href{../ocd/logic1.ps#true}{{\sf true}}` HTML: `<a href="../ocd/logic1.html#forall">∀</a> X.` `<a href="../ocd/logic1.html#true"><b>true</b></a>`
kohlhase-omdoc_F00310	202	■1.000	`\binom{n}{m}`	$\binom{n}{m}$	in Presentation-MATHML. The first **presentation** element in List-
kohlhase-omdoc_F00311	202	■1.000	`m`	m	ing 19.9 shows a **presentation** element t
kohlhase-omdoc_F00312	204	■0.995	`X_{4}`	X_4	
kohlhase-omdoc_F00313	204	■0.995	`\forall X_{4} . X_{4}=X_{4}`	$\forall X_4.X_4 = X_4$	
kohlhase-omdoc_F00314	205	■0.982	`\forall \prec, X . X \prec X`	$\forall \prec, X.X \prec X$	so the second notation declaration governs this and puts it in to infix position.
kohlhase-omdoc_F00315	205	■1.000	`\prec`	$\prec$	Note that while OMDOC allows to specify this kind of notation declarations,
kohlhase-omdoc_F00316	215	■0.755	`\mathrm{T}_{\mathrm{E}} \mathrm{XBook}`	$\mathrm{T}_{\mathrm{E}} \mathrm{XBook}$	210 21 Exercises
kohlhase-omdoc_F00317	215	■0.891	`{\,p\mid\hbox{`	*(not rendered)*	English description; for example consider '{p|p and p+2 are pri
kohlhase-omdoc_F00318	215	■0.891	`and`	*and*	English description; for example consider '{p|p and p+2 are pri
kohlhase-omdoc_F00319	215	■0.891	`are prime}\,}`	*(not rendered)*	English description; for example consider '{p|p and p+2 are pri
kohlhase-omdoc_F00320	215	■0.468	`\{\,p\mid p`	$\{\,p\mid p$	assuming that <phrase style="font−family:fixed">\mat
kohlhase-omdoc_F00321	215	■0.468	`p+2`	$p+2$	assuming that <phrase style="font−family:fixed">\mat
kohlhase-omdoc_F00322	215	■0.468	`\,}`		assuming that <phrase style="font−family:fixed">\mat
kohlhase-omdoc_F00323	215	■0.405	`\{\,p\mid p\\{\rm and}\p+2\rm\ are\ prime\,}`	*(not rendered)*	kin) within math formulas. Of course it may be best to display a formula like

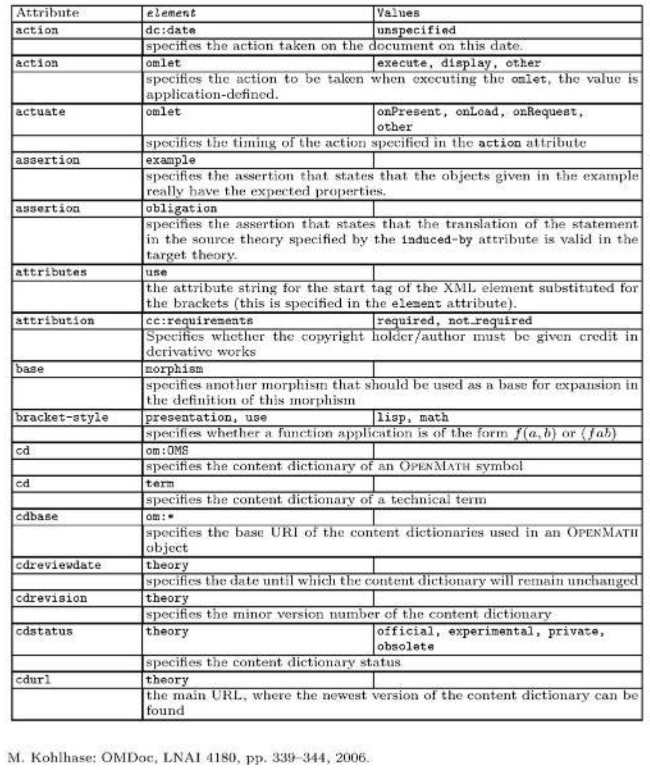
Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_F00324	218	1.000	`\mathcal{K}`	$\mathcal{K}$	tions like e.g. presentation MATHML [ABC+03] or XHTML [Gro00] specify models in form of the intended screen presentation, while still others like the
kohlhase-omdoc_F00325	218	0.935	`{ }^{1} \mathcal{X}`	${ }^{1} \mathcal{X}$	XSLT [Dea99] give the operational semantics.
kohlhase-omdoc_F00326	218	0.998	`\{d \mid d \mathcal{X} d\}=\mathcal{K}`	$\{d \mid d \mathcal{X} d\}=\mathcal{K}$	For a formal definition let $\mathcal{K}$ be a set of documents. We take a **docu-**
kohlhase-omdoc_F00327	218	1.000	`d`	d	that $\{d \mid d\mathcal{X}d\} = \mathcal{K}$. In particular, a relation $\mathcal{X}$ is an equivalence relation on
kohlhase-omdoc_F00328	218	1.000	`d^{\prime}`	$d^{\prime}$	that $\{d \mid d\mathcal{X}d\} = \mathcal{K}$. In particular, a relation $\mathcal{X}$ is an equivalence relation on
kohlhase-omdoc_F00329	218	1.000	`d \mathcal{X} d^{\prime}`	$d \mathcal{X} d^{\prime}$	$\mathcal{K}$. For a given document model $\mathcal{X}$, let us say that two documents d and d'
kohlhase-omdoc_F00330	218	1.000	`p \mathcal{X}`	$p \mathcal{X}$	$\mathcal{K}$. For a given document model $\mathcal{X}$, let us say that two documents d and d'
kohlhase-omdoc_F00331	218	1.000	`d \mathcal{X} d^{\prime}, p`	$d \mathcal{X} d^{\prime}, p$	$\mathcal{K}$. For a given document model $\mathcal{X}$, let us say that two documents d and d'
kohlhase-omdoc_F00332	218	1.000	`p`	p	are $\mathcal{X}$-**equal**, iff $d\mathcal{X}d'$. We call a property holds on d whenever p holds on d'
kohlhase-omdoc_F00333	218	1.000	`[d]_{\mathcal{X}}`	$[d]_{\mathcal{X}}$	
kohlhase-omdoc_F00334	218	0.999	`[d]_{\mathcal{X}}:=\{e \mid d \mathcal{X} e\}`	$[d]_{\mathcal{X}}:=\{e \mid d \mathcal{X} e\}$	
kohlhase-omdoc_F00335	220	1.000	`X+Y=`	$X+Y=$	obtained by exchanging the right hand side $Y+X$ of the equality by $X+Y$,
kohlhase-omdoc_F00336	220	0.268	`Y+X`	$Y+X$	which is mathematically equal (but not OMDoc-equal).
kohlhase-omdoc_F00337	220	1.000	`X+Y=X+Y`	$X+Y=X+Y$	Let us now specify (part of) the equality relation by the rules in the table in Figure 22.4. We have discussed a machine-readable form of these equality
kohlhase-omdoc_F00338	220	1.000	`X+Y`	$X+Y$	constraints in the XML schema for OMDoc in [KA03].
kohlhase-omdoc_F00339	232	0.999	`\left[\mathrm{ABD}^{+} 01\right]`	$\left[\mathrm{ABD}^{+} 01\right]$	**24.1.3 Modularization**
kohlhase-omdoc_F00340	232	0.966	`{ }^{+} 01`	${ }^{+} 01$	driver includes an entity file ⟨⟨*ModId*⟩⟩`.ent` for each module ⟨⟨*ModId*⟩⟩, before
kohlhase-omdoc_F00341	234	0.988	`d\rangle`	$d\rangle$	only be associated with whole documents.
kohlhase-omdoc_F00342	238	1.000	`2 n`	$2 n$	
kohlhase-omdoc_F00343	238	1.000	`n^{2}`	n^{2}	
kohlhase-omdoc_F00344	240	0.970	`\lambda C l a m`	$\lambda C l a m$	
kohlhase-omdoc_F00345	240	0.836	`\left[\mathrm{ABI}^{+} 96\right]`	$\left[\mathrm{ABI}^{+} 96\right]$	
kohlhase-omdoc_F00346	241	0.183	`\mathrm{IAT}_{E} \mathrm{X}`	$\mathrm{IAT}_{E} \mathrm{X}$	first two include the latter and only redefine some format-specific options.
kohlhase-omdoc_F00347	243	0.990	`\mathrm{T}_{\mathrm{E}} \mathrm{X}_{\text {MACS }}`	$\mathrm{T}_{\mathrm{E}} \mathrm{X}_{\text {MACS }}$	Section 26.16, the CPoint add-in for MS PowerPoint (Section 26.14).
kohlhase-omdoc_F00348	243	1.000	`\mathrm{ST}_{\mathrm{E}} \mathrm{X}`	$\mathrm{ST}_{\mathrm{E}} \mathrm{X}$	OMDoc that can be generated by the `qmath` parser. STEX defines a
kohlhase-omdoc_F00349	243	0.220	`_{\text {E }}`	$_{\text {E }}$	
kohlhase-omdoc_F00350	246	1.000	`2+2=4`	$2+2=4$	Theory:[<−thoughtcrime] Syml

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_F00351	251	0.429	`\$ \mathrm{e}^{\wedge}(\mathrm{pi} * \mathrm{i})+1=0 \$`	$\$\mathrm{e}^{\wedge}(\mathrm{pi} * \mathrm{i})+1=0\$$	display updated from the OpenMath form, at each point when there is a
kohlhase-omdoc_F00352	251	1.000	`e^{\pi i}+1=0`	$e^{\pi i}+1=0$	complete parse of the formula. This gives immediate feedback on how the
kohlhase-omdoc_F00353	251	0.370	`{ }^{\mathrm{TM}}`	${ }^{\mathrm{TM}}$	
kohlhase-omdoc_F00354	251	0.370	`{ }^{(\mathrm{R})}`	${ }^{(\mathrm{R})}$	
kohlhase-omdoc_F00355	254	1.000	`O`	O	server S'; a query to O that needs information about O' will be delegated to
kohlhase-omdoc_F00356	254	1.000	`O^{\prime}`	$O^{\prime}$	a suitable query to the server S'. We distinguish two kinds of MBase servers
kohlhase-omdoc_F00357	254	1.000	`S^{\prime}`	$S^{\prime}$	ferred to by other MBases, and *scratch-pad* MBases that are not referred to.
kohlhase-omdoc_F00358	255	1.000	`\binom{n}{k},{ }_{n} C^{k}, C_{k}^{n}`	$\binom{n}{k},{ }_{n} C^{k}, C_{k}^{n}$	2. *Identical presentations can stand for multiple distinct mathematical ob-*
kohlhase-omdoc_F00359	255	1.000	`C_{n}^{k}`	C_{n}^{k}	2. *Identical presentations can stand for multiple distinct mathematical ob-*
kohlhase-omdoc_F00360	255	1.000	`\frac{n!}{k!(n-k)!}`	$\frac{n!}{k!(n-k)!}$	2. *Identical presentations can stand for multiple distinct mathematical ob-*
kohlhase-omdoc_F00361	255	0.999	`\int f(x) d x`	$\int f(x) d x$	to the particular integral type we are interested in at the moment.
kohlhase-omdoc_F00362	255	1.000	`\int f(y) d y`	$\int f(y) d y$	disambiguate mathematical notions.
kohlhase-omdoc_F00363	255	1.000	`s(t)`	$s(t)$	
kohlhase-omdoc_F00364	255	1.000	`\int_{?}^{?} s^{2}(t) d t`	$\int_{?}^{?} s^{2}(t) d t$	
kohlhase-omdoc_F00365	255	1.000	`\frac{1}{T} \int_{0}^{T} s^{2}(t) d t=\Sigma_{k=-\infty}^{\infty}\left|c_{k}\right|^{2}`	$\frac{1}{T} \int_{0}^{T} s^{2}(t) d t=\Sigma_{k=-\infty}^{\infty}\left\|c_{k}\right\|^{2}$	
kohlhase-omdoc_F00366	255	1.000	`c_{k}`	c_{k}	
kohlhase-omdoc_F00367	257	1.000	`A=B`	$A=B$	tation we are using for terms (prefix notation). The mq:generic attribute takes string values and can be specified for any element in the query, making
kohlhase-omdoc_F00368	257	0.737	`B=A`	$B=A$	it into a (named) query variable: its contents are ignored and it matches any
kohlhase-omdoc_F00369	258	0.994	`\frac{1}{T} \int_{0}^{T} s^{2}(x) d x=\Sigma_{k=-\infty}^{\infty}\left|c_{k}\right|^{2}`	$\frac{1}{T} \int_{0}^{T} s^{2}(x) d x=\Sigma_{k=-\infty}^{\infty}\left\|c_{k}\right\|^{2}$	
kohlhase-omdoc_F00370	259	0.319	`\left.{ }^{+} 03, \mathrm{KBLL}{ }^{+} 04, \mathrm{KBKB}^{+} 04\right]`	$\left.{ }^{+} 03, \mathrm{KBLL}{ }^{+} 04, \mathrm{KBKB}^{+} 04\right]$	MMiSSLaTeX package for ontologies provides a set
kohlhase-omdoc_F00371	261	1.000	`T`	T	the rules. There may be good reasons, for instance, to present first a "light-
kohlhase-omdoc_F00372	261	1.000	`A, B`	A, B	the rules. There may be good reasons, for instance, to present first a "light-
kohlhase-omdoc_F00373	261	0.533	`P \& A`	$P \& A$	
kohlhase-omdoc_F00374	261	0.533	`T \& P`	$T \& P$	
kohlhase-omdoc_F00375	261	0.533	`T \Rightarrow B`	$T \Rightarrow B$	
kohlhase-omdoc_F00376	264	0.997	`\mathbb{F}`	$\mathbb{F}$	DocBook type grammar sees to it that there are natural scopes, where math-

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_F00377	264	1.000	`\frac{3 / 4+2 / 3}{5}`	$\frac{3 / 4+2 / 3}{5}$	
kohlhase-omdoc_F00378	264	0.998	`3,4 \in \mathbb{Z}`	$3,4 \in \mathbb{Z}$	
kohlhase-omdoc_F00379	264	1.000	`3 \in \mathbb{Z} \wedge 4 \in \mathbb{Z}`	$3 \in \mathbb{Z} \wedge 4 \in \mathbb{Z}$	
kohlhase-omdoc_F00380	264	0.998	`\mathbb{Z}`	$\mathbb{Z}$	
kohlhase-omdoc_F00381	268	0.608	`\left[\mathrm{MGH}^{+} 05\right]`	$\left[\mathrm{MGH}^{+} 05\right]$	
kohlhase-omdoc_F00382	269	0.995	`\left[\mathrm{KAB}^{+} 04\right]`	$\left[\mathrm{KAB}^{+} 04\right]$	was in first versions of LOM. Metadata values, such as `difficulty`,
kohlhase-omdoc_F00383	273	0.996	`10^{\prime} 000`	$10^{\prime} 000$	the art of properly choosing mathematical semantic encoding: the mathemat-
kohlhase-omdoc_F00384	273	1.000	`1, \ldots, k, \ldots, n`	$1, \ldots, k, \ldots, n$	
kohlhase-omdoc_F00385	275	0.998	`\left[\mathrm{SBB}^{+} 06\right]`	$\left[\mathrm{SBB}^{+} 06\right]$	learning [SBB+00].
kohlhase-omdoc_F00386	279	1.000	`{ }^{\sim}`	${ }^{\sim}$	278 26 Applications and Projects **26.11 Induction Challenge OMDoc Manager (ICOM)** Project Home: `http://www.cs.nott.ac.uk/~lad/research/challenges/challenge_manager.html` Authors: Thomas D. Attfield, Monica C. Duarte, Lin Li, Ho-Ying Mak, Adam M. Neal, Lewis M. Toft, Zixuan Wang, Louise A. Dennis
kohlhase-omdoc_F00387	282	1.000	`{ }^{1} { }^{1}`	${ }^{1} { }^{1}$	26.12 MAYA 281 **26.12 Maya: Maintaining Structured Developments** Project Home: `www.dfki.de/~inka/maya.html` Authors: Serge Autexier[1], Dieter Hutter[1], Till Mossakowski[2], Axel Schairer[1]
kohlhase-omdoc_F00388	282	0.757	`{ }^{+} 00`	${ }^{+} 00$	SL [AHL+00]. All these specification languages provide constructs similar
kohlhase-omdoc_F00389	286	0.830	`\left[\mathrm{SHB}^{+} 99\right]`	$\left[\mathrm{SHB}^{+} 99\right]$	with a treatment of hiding [MAH06], and a uniform treatment of different
kohlhase-omdoc_F00390	295	0.465	`\mathrm{St}_{\mathrm{E}} \mathrm{X}`	$\mathrm{St}_{\mathrm{E}} \mathrm{X}$	in the form of TEX/LATEX documents.
kohlhase-omdoc_F00391	295	0.962	`\mathrm{ST}_{\mathrm{E}}^{\mathrm{X}}`	$\mathrm{ST}_{\mathrm{E}}^{\mathrm{X}}$	documents can be authored and maintained in sTEX using a simple text ed-
kohlhase-omdoc_F00392	295	0.580	`\mathrm{T}_{\mathrm{E}} \mathrm{X} / \mathrm{IAT}_{\mathrm{E}} \mathrm{X}`	$\mathrm{T}_{\mathrm{E}} \mathrm{X} / \mathrm{IAT}_{\mathrm{E}} \mathrm{X}$	
kohlhase-omdoc_F00393	297	0.713	`\binom{n}{k},{ }_{n} C^{k}, \mathcal{C}_{k}^{n}`	$\binom{n}{k},{ }_{n} C^{k}, \mathcal{C}_{k}^{n}$	presentation (i.e. what formulae look like) and not their content (their func-
kohlhase-omdoc_F00394	297	0.713	`\mathcal{C}_{n}^{k}`	$\mathcal{C}_{n}^{k}$	presentation (i.e. what formulae look like) and not their content (their func-
kohlhase-omdoc_F00395	297	1.000	`\sum_{k=1}^{\infty} x^{k}`	$\sum_{k=1}^{\infty} x^{k}$	(*i*) the bound variable k (*ii*) the number 1 (*iii*) ∞ (*iv*) `\Cexp{x}k` (i.e. x to
kohlhase-omdoc_F00396	297	0.943	`\infty`	∞	superscript is ∞. Then it should place next to it an x with an upper index k.
kohlhase-omdoc_F00397	297	0.870	`k=1`	$k=1$	tent structure from the expression $\sum_{k=1}^{\infty} x^{k}$ of the right-hand representation

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_ F00398	298	0.996	`\sum_{i=1}^{n} x^{i}`	$\sum_{i=1}^{n} x^{i}$	arbitrary (but fixed) symbol (compare with the use of \arbitraryn below,
kohlhase-omdoc_ F00399	298	1.000	`\arbitraryn`	*(not rendered)*	What is the sum of the first n odd numbers, i.e. $\sum_{i=1} 2i-1$?
kohlhase-omdoc_ F00400	298	1.000	`\Sumfromto{i}1\arbitraryn{2i-1}?`	*(not rendered)*	Moreover, the semantic macro Sumfrom
kohlhase-omdoc_ F00401	298	0.998	`\sum_{i=1}^{n} 2 i-1`	$\sum_{i=1}^{n} 2i-1$	The sTeX macro packages have been validated together with a case
kohlhase-omdoc_ F00402	298	0.660	`\mathrm{S}_{\mathrm{E}} \mathrm{X}`	$\mathrm{S}_{\mathrm{E}} \mathrm{X}$	
kohlhase-omdoc_ F00403	303	0.999	`a x^{2}+b x+c`	$ax^{2}+bx+c$	
kohlhase-omdoc_ F00404	306	0.994	`\Omega_{\text {MEGA }}`	Ω_{MEGA}	26.18 Standardizing Context in System Interoperability 305 PVS / ΩMEGA ‖ PVS / ΩMEGA set / ‖ subset? / subset member / in ‖ / subset2 empty? / empty ‖ strict_subset? / proper-subset emptyset / emptyset ‖ / superset nonempty? / not-empty ‖ union / union full? / ‖ / union2 fullset / ‖ / union-over-collection singleton? / singleton ‖ intersection / intersection singleton / ‖ / intersection-over-coll. complement / set-complement ‖ disjoint? / misses
kohlhase-omdoc_ F00405	306	1.000	`\rho_{*}`	ρ_{*}	the integration theory, using the partial inverse ρ_*^{-1} of ρ_*. For an integration of a set of software systems this refactoring process is repeated recursively
kohlhase-omdoc_ F00406	306	1.000	`\rho_{*}^{-1}`	ρ_{*}^{-1}	of the mathematical software systems, we only rationally reconstruct their meaning in terms of the new integration theory, which can act as a gold stan-
kohlhase-omdoc_ F00407	308	1.000	`\mathcal{S}=(\mathcal{L}, \mathcal{C})`	$\mathcal{S}=(\mathcal{L}, \mathcal{C})$	
kohlhase-omdoc_ F00408	308	1.000	`\mathcal{L}`	$\mathcal{L}$	
kohlhase-omdoc_ F00409	308	0.999	`\mathcal{H}`	$\mathcal{H}$	
kohlhase-omdoc_ F00410	309	1.000	`\mathcal{D}: \mathcal{H} \vdash_{\mathcal{C}} \mathbf{A}`	$\mathcal{D}: \mathcal{H} \vdash_{\mathcal{C}} \mathbf{A}$	
kohlhase-omdoc_ F00411	309	1.000	`\mathcal{S}^{\prime}`	$\mathcal{S}^{\prime}$	
kohlhase-omdoc_ F00412	309	1.000	`\mathcal{F}: \mathcal{S} \rightarrow \mathcal{S}^{\prime}`	$\mathcal{F}: \mathcal{S} \rightarrow \mathcal{S}^{\prime}$	
kohlhase-omdoc_ F00413	309	1.000	`\mathcal{F}^{\mathcal{L}}: \mathcal{L} \rightarrow \mathcal{L}^{\prime}`	$\mathcal{F}^{\mathcal{L}}: \mathcal{L} \rightarrow \mathcal{L}^{\prime}$	
kohlhase-omdoc_ F00414	309	1.000	`\mathcal{F}^{\mathcal{D}}`	$\mathcal{F}^{\mathcal{D}}$	308 26 Applications and Projects
kohlhase-omdoc_ F00415	309	1.000	`\mathcal{C}^{\prime}`	$\mathcal{C}^{\prime}$	308 26 Applications and Projects
kohlhase-omdoc_ F00416	309	0.999	`\mathcal{F}^{\mathcal{D}}(\mathcal{D}): \mathcal{F}^{\mathcal{L}}(\mathcal{H}) \vdash_{\mathcal{C}^{\prime}} \mathcal{F}^{\mathcal{L}}(\mathbf{A})`	$\mathcal{F}^{\mathcal{D}}(\mathcal{D}): \mathcal{F}^{\mathcal{L}}(\mathcal{H}) \vdash_{\mathcal{C}^{\prime}} \mathcal{F}^{\mathcal{L}}(\mathbf{A})$	308 26 Applications and Projects
kohlhase-omdoc_ F00417	309	1.000	`\mathcal{R}`	$\mathcal{R}$	The intuition behind this is that logic morphisms transport proofs between
kohlhase-omdoc_ F00418	309	1.000	`(\mathbb{S} F O L)`	$(\mathbb{S} FOL)$	The intuition behind this is that logic morphisms transport proofs between
kohlhase-omdoc_ F00419	309	1.000	`(F O L)`	(FOL)	logical systems. Logic morphisms come in all shapes and sizes, a well-known one is the relativization morphism from sorted logics to unsorted ones, for

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_F00420	309	1.000	\mathcal{R}([+: \mathbb{N} \rightarrow \mathbb{N} \rightarrow \mathbb{N}])=\forall X, Y . \mathbb{N}(X) \wedge \mathbb{N}(Y) \Rightarrow \mathbb{N}(X+Y)	$\mathcal{R}([+: \mathbb{N} \rightarrow \mathbb{N} \rightarrow \mathbb{N}])=\forall X, Y . \mathbb{N}(X) \wedge \mathbb{N}(Y) \Rightarrow \mathbb{N}(X+Y)$	instance the morphism $\mathcal{R}$ from sorted first-order logic ($\mathbb{S}FOL$) to unsorted
kohlhase-omdoc_F00421	309	0.999	\mathcal{R}\left(\forall X_{\mathbb{B}} . \mathbf{A}\right)=\forall X . \mathbb{B}(X) \Rightarrow \mathcal{R}(\mathbf{A})	$\mathcal{R}\left(\forall X_{\mathbb{B}} . \mathbf{A}\right)=\forall X . \mathbb{B}(X) \Rightarrow \mathcal{R}(\mathbf{A})$	lae sorted quantifications are translated into unsorted ones guarded by sort
kohlhase-omdoc_F00422	309	1.000	\mathbb{A}, \mathbb{B}, \ldots	$\mathbb{A}, \mathbb{B}, \ldots$	predicates, e.g. $\mathcal{R}(\forall X_{\mathbb{B}}.\mathbf{A}) = \forall X.\mathbb{B}(X) \Rightarrow \mathcal{R}(\mathbf{A})$. Finally, for proofs we have
kohlhase-omdoc_F00423	309	0.989	\mathbb{S} F O L	$\mathbb{S} F O L$	
kohlhase-omdoc_F00424	309	0.989	F O L	$F O L$	
kohlhase-omdoc_F00425	310	0.752	\left[\mathrm{SBB}^{+} 02\right.	$\left[\mathrm{SBB}^{+} 02\right.$	professional typesetting and supports authoring with powerful macro defi-
kohlhase-omdoc_F00426	310	0.975	\mathrm{T}_{\mathrm{E}} \mathrm{X}_{\text {MACS }}[\mathrm{dH} 01]	$\mathrm{T}_{\mathrm{E}} \mathrm{X}_{\text {MACS }}[\mathrm{dH} 01]$	definable macros. The full access to the document format together with the
kohlhase-omdoc_F00427	312	1.000	T+S	$T+S$	we define a **semantic document format** $T+S$ as a document format still
kohlhase-omdoc_F00428	312	1.000	S(P)	$S(P)$	typeset with $\mathrm{T_EX_{MACS}}$.
kohlhase-omdoc_F00429	312	1.000	S(C)	$S(C)$	The $\mathrm{T_EX_{MACS}}$ document $T+S(C)$ and the pure semantic representation
kohlhase-omdoc_F00430	312	1.000	C	C	$S(C)$ in the proof assistant must be synchronized. The basic idea here is to
kohlhase-omdoc_F00431	312	0.993	T+S(C)	$T+S(C)$	to s_i results in s_{i+1} which corresponds to ts_{i+1}. An analogous diff/patch
kohlhase-omdoc_F00432	312	0.993	t s_{i}	$t s_{i}$	used to identify the corresponding parts in $T+S(C)$ and $S(C)$.
kohlhase-omdoc_F00433	312	0.993	t s_{i+1}	$t s_{i+1}$	used to identify the corresponding parts in $T+S(C)$ and $S(C)$.
kohlhase-omdoc_F00434	312	1.000	s_{i}	s_{i}	proof assistant. This language M is a language for structured menus and ac-
kohlhase-omdoc_F00435	312	1.000	s_{i+1}	s_{i+1}	
kohlhase-omdoc_F00436	313	1.000	T+S(C)+M	$T+S(C)+M$	
kohlhase-omdoc_F00437	313	1.000	S(C)+M	$S(C)+M$	with the proof assistants. The $\mathrm{T_EX_{MACS}}$ document format $T+S(C)$ is finally
kohlhase-omdoc_F00438	314	1.000	\mathcal{F} \mathcal{P}	$\mathcal{F} \mathcal{P}$	
kohlhase-omdoc_F00439	316	0.537	h_{1}, \ldots h_{n}, \forall \ldots i h_{1}, \ldots, \forall \ldots i h_{l} \vdash	$h_{1}, \ldots h_{n}, \forall \ldots i h_{1}, \ldots, \forall \ldots i h_{l} \vdash$	
kohlhase-omdoc_F00440	316	1.000	h_{i}	h_{i}	method elements. Parameters heuristically computed by the system or manu-
kohlhase-omdoc_F00441	316	1.000	\forall \ldots i h_{k}	$\forall \ldots i h_{k}$	method elements. Parameters heuristically computed by the system or manu-
kohlhase-omdoc_F00442	332	—	\begin{aligned} & ([0-9]|,| .) \\ & (* \mid \mathrm{e}([0-9]|,| .) *) ? \end{aligned}	$([0-9] \vert , \vert .)$ $(* \mid \mathrm{e}([0-9] \vert , \vert .) *) ?$	—

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_ F00443	338	0.805	`f(a, b)`	$f(a, b)$	

Attribute	*element*	Values
action	dc:date	unspecified
	specifies the action taken on the document on this date.	
action	omlet	execute, display, other
	specifies the action to be taken when executing the omlet, the value is application-defined.	
actuate	omlet	onPresent, onLoad, onRequest, other
	specifies the timing of the action specified in the action attribute	
assertion	example	
	specifies the assertion that states that the objects given in the example really have the expected properties.	
assertion	obligation	
	specifies the assertion that states that the translation of the statement in the source theory specified by the induced-by attribute is valid in the target theory.	
attributes	use	
	the attribute string for the start tag of the XML element substituted for the brackets (this is specified in the element attribute).	
attribution	cc:requirements	required, not_required
	Specifies whether the copyright holder/author must be given credit in derivative works	
base	morphism	
	specifies another morphism that should be used as a base for expansion in the definition of this morphism	
bracket-style	presentation, use	lisp, math
	specifies whether a function application is of the form $f(a,b)$ or $(f a b)$	
cd	om:OMS	
	specifies the content dictionary of an OpenMath symbol	
cd	term	
	specifies the content dictionary of a technical term	
cdbase	om:*	
	specifies the base URI of the content dictionaries used in an OpenMath object	
cdreviewdate	theory	
	specifies the date until which the content dictionary will remain unchanged	
cdrevision	theory	
	specifies the minor version number of the content dictionary	
cdstatus	theory	official, experimental, private, obsolete
	specifies the content dictionary status	
cdurl	theory	
	the main URL, where the newest version of the content dictionary can be found	

M. Kohlhase: OMDoc, LNAI 4180, pp. 339–344, 2006.

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Image</th></tr>
<tr><td>kohlhase-omdoc_F00444</td><td>338</td><td>0.805</td><td><code>f a b</code></td><td>fab</td><td>
<table>
<tr><td>Attribute</td><td>element</td><td>Values</td></tr>
<tr><td>action</td><td>dc:date</td><td>unspecified</td></tr>
<tr><td></td><td colspan="2">specifies the action taken on the document on this date.</td></tr>
<tr><td>action</td><td>omlet</td><td>execute, display, other</td></tr>
<tr><td></td><td colspan="2">specifies the action to be taken when executing the omlet, the value is application-defined.</td></tr>
<tr><td>actuate</td><td>omlet</td><td>onPresent, onLoad, onRequest, other</td></tr>
<tr><td></td><td colspan="2">specifies the timing of the action specified in the action attribute</td></tr>
<tr><td>assertion</td><td>example</td><td></td></tr>
<tr><td></td><td colspan="2">specifies the assertion that states that the objects given in the example really have the expected properties.</td></tr>
<tr><td>assertion</td><td>obligation</td><td></td></tr>
<tr><td></td><td colspan="2">specifies the assertion that states that the translation of the statement in the source theory specified by the induced-by attribute is valid in the target theory.</td></tr>
<tr><td>attributes</td><td>use</td><td></td></tr>
<tr><td></td><td colspan="2">the attribute string for the start tag of the XML element substituted for the brackets (this is specified in the element attribute).</td></tr>
<tr><td>attribution</td><td>cc:requirements</td><td>required, not_required</td></tr>
<tr><td></td><td colspan="2">Specifies whether the copyright holder/author must be given credit in derivative works</td></tr>
<tr><td>base</td><td>morphism</td><td></td></tr>
<tr><td></td><td colspan="2">specifies another morphism that should be used as a base for expansion in the definition of this morphism</td></tr>
<tr><td>bracket-style</td><td>presentation, use</td><td>lisp, math</td></tr>
<tr><td></td><td colspan="2">specifies whether a function application is of the form $f(a,b)$ or (fab)</td></tr>
<tr><td>cd</td><td>om:OMS</td><td></td></tr>
<tr><td></td><td colspan="2">specifies the content dictionary of an OpenMath symbol</td></tr>
<tr><td>cd</td><td>term</td><td></td></tr>
<tr><td></td><td colspan="2">specifies the content dictionary of a technical term</td></tr>
<tr><td>cdbase</td><td>om:*</td><td></td></tr>
<tr><td></td><td colspan="2">specifies the base URI of the content dictionaries used in an OpenMath object</td></tr>
<tr><td>cdreviewdate</td><td>theory</td><td></td></tr>
<tr><td></td><td colspan="2">specifies the date until which the content dictionary will remain unchanged</td></tr>
<tr><td>cdrevision</td><td>theory</td><td></td></tr>
<tr><td></td><td colspan="2">specifies the minor version number of the content dictionary</td></tr>
<tr><td>cdstatus</td><td>theory</td><td>official, experimental, private, obsolete</td></tr>
<tr><td></td><td colspan="2">specifies the content dictionary status</td></tr>
<tr><td>cdurl</td><td>theory</td><td></td></tr>
<tr><td></td><td colspan="2">the main URL, where the newest version of the content dictionary can be found</td></tr>
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M. Kohlhase: OMDoc, LNAI 4180, pp. 339–344, 2006.
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